Class 8

Physics

Volume 2

Solutions

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5. WORK, POWER AND ENERGY

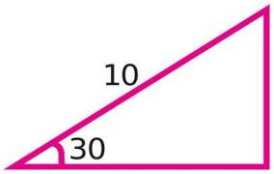
WORKSHEET – 1
Level – I

1. D	3. B	5. A	7. D	9. C
2. C	4. D	6. D	8. D	10. C

Level – II

1) C	2) C	3) B	4) D	5) C	6) C	7) B
8) C	9) B	10) A	11) D	12) B	13) B	14) B
15) C	16) B	17) B	18) B,C	19) A	20) D	21) B
22) C	23) C	24) A 25) A,B		26) 6J 27) a-p; b-q; c-r,s; d-t		
28) C	29) A	30) B				

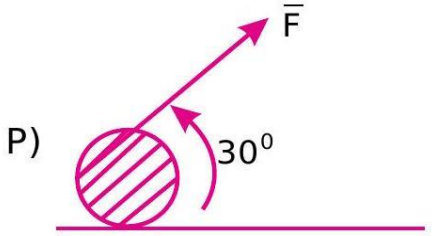
1. $I\sin\theta = 10\sin\theta = 10\sin 30 = 5$ m $h = l\sin\theta$



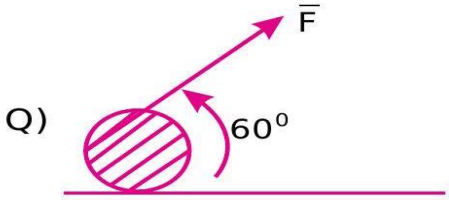
$W = mgh = 5 \times 10 \times 5 = 250$ J

2. Weight and motion are perpendicular to each other
So work done is zero

3. $W = F \times S = 100 \times 4 = 400$ J (Negative)

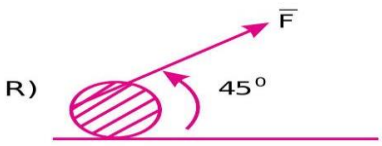


$$W = FS\cos 30^\circ = \frac{\sqrt{3}}{2}FS$$



$W = FS\cos 60^\circ$

$= \frac{1}{2}FS$



$W = FS\cos 45^\circ$

$= \frac{1}{\sqrt{2}}FS$

S)



$W = FS$

$W_{max} = FS$

5. Work done is zero

6. Work done by force $= 40 \times 2 = 80$ N(+ve)
Work done by gravity $= mgh$
$= 2 \times 10 \times 2 = 40(-Ve)$

7. $W = F \cdot S = (4i + 1j + 7k) \cdot (1i + 2j + 1k)$
$= (4 \times 1) + (1 \times 2) + (7 \times 1)$
$= 4 + 2 + 7 = 13$ J

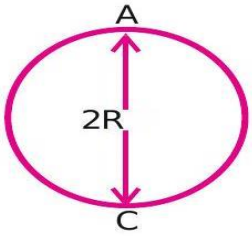
8. $W = mgh = \frac{1}{2} \times 10 \times 1.6 = 8$ J

9. Work done is equal to area under F-s graph.
$$W = \frac{1}{2} \times (14 - 2) \times 60 = 6 \times 60 = 360 \text{ J}$$

10. $W = FS\cos\theta = 16 \times 2.2 \times \cos 45°$
$$= 16 \times 2.2 \times \frac{1}{\sqrt{2}}$$
$$= 16 \times 2.2 \times \frac{1}{1.414} = 24.9;$$
Key : 1

11. Friction $= \mu mg = 0.5 \times 5 \times 10 = 25$ N
$W = F.S = 25 \times 0.1 = 2.5$ J (negative)
Force is equal to friction
So work done by force is also 2.5J

12. $W = mgh = mg \times 2R = 2Rmg$



13. $F = m(a + g) = 5(2 + 10) = 60$ N
Work done by force $= F \times S$
$$= 60 \times 2.5 = 150 \text{ J}$$
Tension $= m(a + g)$
So work done = 150 J positive Work done by gravity $= 5 \times 10 \times 2.5 = 125$ J (negative)

14. In horizontal direction work done is zero
In vertical direction work done $= 10 \times 10 = 100$ J

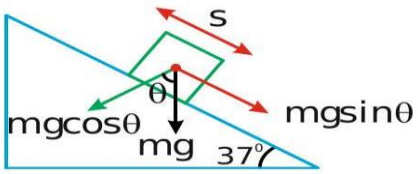
15. $W = (mg - \rho m) \times h$
$$(400 - 40 \times 6) \times 10 = (400 - 240) \times 10$$
$$160 \times 10 = 1600 \text{ J}$$

16. Work is independent of time
So B lifted more height more work done by ' B '

17. $W = F\cos\theta = 100 \times 10 \times \cos 45^\circ$
$= 100 \times 10 \times$
$100 \times 10 \times \frac{1}{\sqrt{2}} = 100 \times 10 \times \frac{1}{1.414} = 707.1$ J
18. Conceptual
19. If a particle moves along a circumference of the circle the work done by the centripetal force is zero. Because centripetal force and tangential motion of the body perpendicular to each other.
20. The work done by a centripetal force is always Zero. Here force and displacement are perpendicular to each other. ($\theta = 90^\circ$)
(or)
The work done by the centripetal force is always zero.
A, B, C are wrong: The magnitude of centripetal force is constant at any point on the
circular path, it may be increased (or) decreased the radius of the circle & mass does not changed.
21.

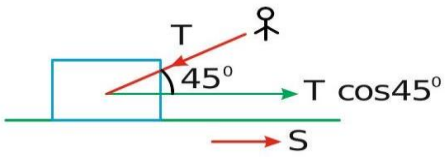


resisting force
$W = FS\cos 180$
$W = -ve$
$W = \vec{F} \cdot \vec{S} = FS\cos\theta = (100 \times 4)\cos 180^\circ$
$= 400\cos(90 + 90) = 400(-\sin 90)$
$= -400(1) = -400$ (− indicates work done by resisting force opposite direction)
22. $w = \vec{F} \cdot \vec{S} = 0 \Rightarrow (3\hat{\imath} + 4\hat{\jmath}) \cdot (4\hat{\imath} + K\hat{\jmath}) = 0$
$12 + 4K = 0; K = -3$
23. $W = F.S$
$= mas$
$= 2 \times 0.5 \times 40$
$= 2 \times \frac{1}{2} \times 40$
$= 40$ J
24. $3y + kx = 5$
$\frac{y}{5/3} + \frac{x}{5/k} = 1$
$\bar{r_1} = \frac{5}{3}\hat{\jmath}$
$\bar{r_2} = \frac{5}{k}\hat{\imath}$
$\bar{S} = \bar{r_2} - \bar{r_1}$
$\bar{s} = \frac{5}{k}\hat{\imath} - \frac{5}{3}\hat{\jmath}$
$\bar{F} \cdot \bar{S} = 0$
$k = 2$
25. Answer 1 and 2
26. $W = mgh = 0.2 \times 10 \times 3 = 6$ J
27. $w = \bar{F} \cdot \bar{S} = (5\hat{\imath} + 5\hat{\jmath}) \cdot (\hat{\imath} - \hat{\jmath}) = 0$ J



$w = mgsin\,\theta \times s$

30.



$w = (T\cos\theta)S$

$= 100 \times \frac{1}{\sqrt{2}} \times 10 = \frac{1000}{1.414} = 707.1\ J$

WORKSHEET – 2

Level – I

1. C
2. D
3. D
4. A
5. D
6. A
7. B
8. D

Level – II

1) C	2) C	3) A	4) B	5) C	6) C	7) D
8) B	9) B	10) B	11) B	12) B	13) B	14) A
15) C	16) B	17) A	18) B	19) C	20) C	21) B,C
22) B	23) A	24) C	25) D	26) B	27) C	28) B
29) B	30) C	31) C	32) A	33) 7	34) C	

2. $W = \frac{1}{2}\ K(x_1^2 - x_2^2)$

$= \frac{1}{2} \times 8((15)^2 - (5)^2)$

$= 4(225 - 25)$

$= 4 \times 200 = 800 = 8\ J$

3. $W = \frac{1}{2}\ K \times (0.04)^2 = 2$

$K \times 0.0016 = 4$

$K = \frac{40000}{16}$

$F = \frac{40000}{16} \times 0.08 = 200N$

4. $P.E = \frac{1}{2}Kx^2 = 50x^2$

then $K = 100\frac{N}{m^2}$

5. $\frac{1}{2} \times K \times (0.2)^2 = 1$

$K.\ (0.04) = 2$

$K = \frac{200}{4} = 50\frac{N}{m}$

6. $W = \frac{1}{2} \times 8(0.15^2 - 0.1^2)$

$= 4(0.0225 - 0.01)$
$= 4(0.0125)$
$= 0.0500$

7. $W = mgh = 10 \times 9.8 \times 20$
$98 \times 20 = 1960$ J

8. $W = \frac{1}{2}\ Kx^2 = \frac{1}{2} \times 100 \times (0.1)^2$

$= 50 \times 0.01 = 0.5 = \frac{1}{2}$ J

9. $\frac{F_1}{F_2} = \frac{q_1 q_2}{(2)^2} \times K \times \frac{(4)^2}{q_1 q_2 K} = \frac{4}{1}$

10. $W = \left(40 + \frac{20}{2}\right) \times 9.8 \times 40$

$= 19600$ J

11. $W = 20 \times 9.8 \times 2$
40×9.8
$4 \times 98 = 392J$

12. $h = ut - \frac{1}{2}gtt^2$

$= 20 \times 2 - \frac{1}{2}10 \times (2)^2$

$40 - 5 \times 4 = 20$
$E = 2 \times 10 \times 20 = 400$ J

13. $W = \frac{1}{2}\ K(x_1^2 - x_2^2)$

$= \frac{1}{2} \times 800(0.16 - 0.08^2)$
$= 400(0.0256 - 0.0064)$
$= 7.68$ J (negative)

14. mgh

$= (55 + 25) \times 10 \times 20$
$= 1600$ J $= 16$ kJ

15. $U_1 = \frac{1}{2}\ Kx^2$

$U_2 = \frac{1}{2}K(x + y)^2$

$U_2 - U_1 = \frac{1}{2}K[x^2 + y^2 + 2xy - x^2]$

$\frac{1}{2}Ky(y + 2x)$

16. Energy $= \frac{1}{2}mv^2 = \frac{1}{2} \times (2)^2 = 8$ J

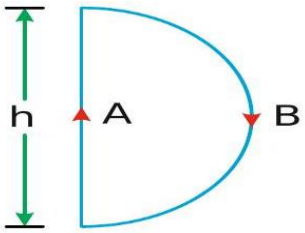
$8 = \frac{1}{2} \times 400 \times x^2$

$x = \sqrt{\frac{16}{400}} = \frac{4}{20} = 0.2$ m

17. $W = mgh$

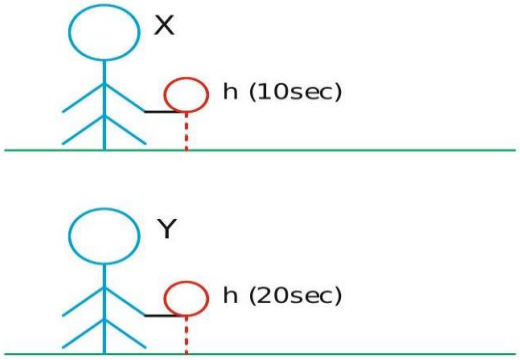
$= 100 \times 9.8 \times 40 = 39200$ J

18. Conceptual



When a body lifted from the ground to a certain height along two paths then, here equal work is done. So, there is no change in force.
(or)
Work done by the conservative force, so this force depends on only initial and final position of the path, Independent of the path followed
A, B, D are wrong:
Work done by the conservative force not path followed

20.



If X person the same mass raised to same height but in 10sec and if Y person this same mass raised to same height in 20sec
$W = F.S$
$W = ma \times S$ a = acceleration
S = displacement
W = work done
(or)
Work depends on mass and distance, so these are same for two persons.
Hence work done by X and Y are same

21. Energy units $kgm^2\ s^{-2}$
Force: $kgms^{-2}$ (or) newton
$kgms^{-2}$(m)
S.I of energy: newton - meter
A, D are wrong: erg is C.G.S unit of energy

22. Work done by the conservative force is Independent of the path followed it depends on Initial and final position. So work done by Gravitational force along paths 1,2,3 are W_1, W_2, W_3 respectively.
Hence work done for conservative force $W_1 = W_2 = W_3$
A, C, D are wrong. Conservative force is Independent of path taken.

23. Comet move around the sun in elliptical orbit, Attractive force between comet and sun is gravitational force.
The work done by gravitational force over a complete orbit is zero. So gravitational force is a conservative force.
Comet move around the sun in elliptical orbit, Attractive force between comet and sun is gravitational force. The work done by gravitational force over a complete orbit is zero. So gravitational force is a conservative force.

24. Work done by Gravitational force is Independent of path so gravitational force is a conservation force.
25. Work done by frictional force through round trip is not zero. But here given to zero so assertion is false.
26. The energy possessed by a bent bow is P.E
27. Work done In Gravitational field (By Gravitational force) depends on End positions of the path.
B, C, D are wrong:
Conservative force (Gravitational force) not depends on path distance, Actual path motion, velocity of transport of mass.
28. $W = \int_1^2 F dx = (2x + x^2)|_1^2 = 8$ J
29. $W = \int_{(0,0)}^{(1,1)} \vec{F} \cdot \overrightarrow{ds}$;
Here $\overrightarrow{ds} = dx\hat{\imath} + dy\hat{\jmath} + dz\hat{k}$

$$\therefore W = \int_{(0,0)}^{(1,1)} (x^2 dy + y dx) = \int_{(0,0)}^{(1,1)} (y^2 dy + x \cdot dx)$$

(as $x = y$)

30. $W = \int_0^{X_1} F dx$
31. $W = \int_0^{X_1} F dx$
32. $W = \frac{1}{2}mv^2$ where $v = \frac{dx}{dt}$
33. $W = \frac{1}{2}mv^2$
34. $U = \frac{1}{2}kx^2$

WORKSHEET – 3
Level – I

1. A	3. C	5. B	7. B
2. D	4. B	6. A	8. D

1. When a spring is stretched (or) compressed by an amount form its unstretched position the energy stored in spring is called Elastic Energy.
$U = \frac{1}{2}kx^2$
K = spring constant
For a spring ' k ' is constant so the E.P.E depends on ' x ' [compression (or) elongation]
2. Conceptual
3. $U = \frac{-Gm_1 m_2}{r}$
4. The PE between two electrons system is Distance between two electrons is decreased PE is Increased A, C, D are wrong: PE between two electron system is not zero, If the electrons come closer to each other PE is increased (not decreasing) and not Infinity.
5. $w = \frac{4}{3}\pi r^3 sg \times h$
$w \alpha r^3$
6. $W = \frac{1}{2}kx^2 \Rightarrow x$ constant
So, $W \propto k$
7. $W = \frac{1}{2}Kx^2 = \frac{1}{2}K\frac{F^2}{K^2} = \frac{1}{2}\frac{F^2}{K} \Rightarrow W \alpha \frac{1}{K}$
F = constant

8. Conceptual

Level – II

1) D	2) B	3) A	4) B	5) C	6) D	7) C
8) B	9) D	10) A	11) B	12) B	13) C	14) C
15) D	16) A	17) B	18) C	19) A	20) A	21) C
22) B						

1. $W = mgh = 10 \times 9.8 \times 20$
$= 98 \times 20 = 1960$ J
2. $k = 100$ N/m, $x = 10$ cm
$= 10 \times 10^{-2}$ m
$W = \frac{1}{2}kx^2$
$= \frac{1}{2} \times 100(10 \times 10^{-2})^2 = \frac{1}{2} \times 100 \times 10^{-2} = \frac{1}{2}$ J
3. $P \cdot E_i = U_i = -\frac{GMm}{R}$
(On the surface of earth)
P. $E_f = U_f = -\frac{GMm}{2R}$ (Certain height from the surface of earth)
$W = (\text{P.E.}) = \frac{mgh}{1+\frac{h}{R}} = \frac{mgR}{2}$ [$\because h = R$]
(or)
$W =$ Gain in P.E. $= U_i - (-U_f) = U_i - U_f$
$= GMm\left[\frac{1}{r_1} - \frac{1}{r_2}\right]$
$= GMm\left[\frac{1}{R} - \frac{1}{2R}\right]$
$= \frac{GMm}{2R} = \frac{GM}{R^2}\frac{mR}{2} = \frac{mgR}{2}$
Since $\left(\because g = \frac{GM}{R^2}\right)$
4. $u = \frac{k \cdot q_1 q_2}{r}$
$u \propto \frac{1}{r}$
$\frac{u_1}{u_2} = \frac{r_2}{r_1} = \frac{4}{2} = 2:1$
5. $W = Mgh + \frac{mgh}{2}$
$Mgh =$ Mass of bucket
$Mgh =$ mass of rope
$w\left(M + \frac{m}{2}\right)gh = \left(40 + \frac{20}{2}\right) \times 9.8 \times 40$
$\Rightarrow 50 \times 40 \times 9.8 = 19600$ J
6. $E_{P \cdot E_1} = \frac{1}{2} \cdot K \cdot (1)^2 = \frac{K}{2}$
$E_{P \cdot E_2} = \frac{1}{2} \cdot K \cdot (2)^2 = 4E_{P \cdot E_1}$
7. $\theta = 90°$, then $W = 0$
8. $w = mgh = 20 \times 2 \times 9.8 = 392$ J
9. Elastic Potential energy $= \frac{1}{2}$ F.x $x = 1$ mm

$= 1 \times 10^{-3}$ m

$= \frac{1}{2}(200)(1 \times 10^{-3})$

$= 0.1$ J

(or)

$$w = \left(\frac{F}{2}\right)x = \frac{1}{2} \times 200 \times \frac{1}{1000} = 0.1 \text{ J}$$

10. $W = \frac{1}{2}mgl$ (or) $W = \frac{1}{2}$ F.$x \Rightarrow \frac{1}{2}mgx$

$$= \frac{1}{2} \times 5 \times 10 \times 3 = 75 \text{ J}$$

or

$w = \left(\frac{F}{2}\right)x = \left(\frac{5 \times 10}{2}\right)3 = 75$ J

11. $s = ut + \frac{1}{2}at^2$, $h = ut - \frac{1}{2}gt^2$

$= (20)2 - \frac{1}{2}(10)(\ ^2\!A) = 40 - 20 = 20$ m

After 2sec $PE = mgh = 400$ J

12. $W = \frac{1}{2}k[x_2^2 - x_1^2]$

$= \frac{1}{2} \times 800[(15 \times 10^{-2}) - (5 \times 10^{-2})^2]$

$= 400[225 \times 10^{-4} - 25 \times 10^{-4}]$

$= 400[200 \times 10^{-4}]$

$= 8 \times 10^4 \times 10^{-4} = 8$ J

13. $W_g = -mgsin\,\theta.l$

$$W_g = -mg\frac{h}{l} \cdot l$$

$W_g = -mgh \Rightarrow W_g = mgh$

14. $w = mgh = 6 \times 9.8 \times 2 = 117.6$ J $\simeq 118$ J

15. $w = mgh$

16. P.E $= mgh_1$; here $h_1 = \frac{h}{2}$ and $m = \rho \times V$

17. Heat produced = loss of potential energy = mgh

18. $U = \frac{1}{2}Kx_1^2 \Rightarrow K = \frac{2U}{x_1^2}$ and $F = Kx_2$

19. $U = \frac{1}{2}Kx^2 - -(1), U = 50x^{2--(2)}$

Compare equation (1) & (2) to find K.

WORKSHEET – 4
Level – I

1. A	4. B	6. A	8. A
2. D	5. B	7. A	9. A
3. B			10. B

1. $KE = \frac{1}{2}mv^2$

Here ' M ' and are V^2 always positive then KE is positive. B, C, D are wrong. PE is may be positive (or) Negative (or) zero. When a body is lifted to a certain height from the ground this work is stored in PE.
The work done by lifted force is +ve (positive). The work done by Gravitational force is negative

2. From Newton's III rd law force applied by the body $= -F = \frac{mv^2}{2\,s} \rightarrow (1)$

$F \propto \frac{1}{V}$

$F \propto \frac{1}{s/t} \Rightarrow F \propto \frac{t}{s} \rightarrow (2)$

From (1) and (2)

$\frac{mv^2}{2s} \propto \frac{t}{s}$

$\frac{1}{2}mv^2 \propto t$

$KE \propto t$

∴ Kinetic Energy is directly proporti onal to time

A, C, D are wrong:

If force acting on a body is inversely proportional to speed, then the KE is not constant, not inversely proportional to time not directly proportional square of time.

3. A body at rest can have PE Potential Energy: The Energy possessed by a body by virtue of its position (or) configuration is called *PE*

A, B, C are wrong:

Kinetic Energy - by virtue of motion

Momentum(P)- Mass xvelocity

[At Rest = 0]

$P = 0$

4. The force is directly proportional to the acceleration of the body $F \propto a$

$F = ma$

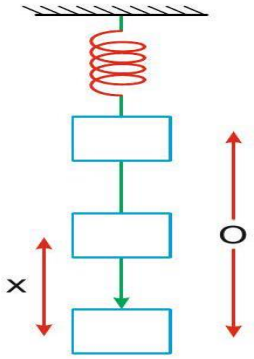
Where m = proportional constant (mass)

$K.E = FS$

$KE \propto S$[∴ Fis constant]

5. If we are applying Retarding forces are equal i. e constant. So applied force is constant. This force is Independent of mass of the bodies. So bus and car covered with Equal distances before coming at rest.

6. The oscillations of spring is S. H. M K. E in S. H. M is



$KE = \frac{1}{2}\ m\omega^2(\ A^2 - x^2)$

At middle $x = 0$

$KE_{max} = \frac{1}{2}\ m\omega^2\ A^2$

KE is maximum at the middle of the moment.

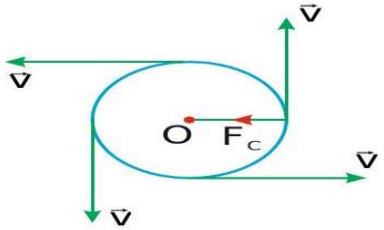
At the bottom Amplitude $(A) = x$

Just before released i.e maximum displacement nothing but a Amplitude

$\therefore x = 0[\mathrm{KE}_{\min} = 0]$
KE is not constant

7. Retarding force is constant. Constant Force is Independent of mass of the bodies. So lorry and car before come to rest travelling in same distance.

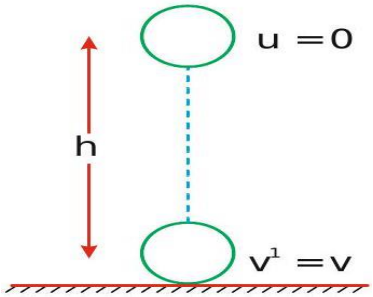
8. If a force is always perpendicular to the motion. We assumption that in uniform circular motion speed is constant



Work done per centripetal force is zero, KE is also remains constant. Velocity is not constant its changes continuously.

9. KE of a particle remains constant in uniform circular motion. In a uniform circular motion, the magnitude of centripetal force is constant, so that KE of a particle not Increased it is remains constant

10. Gravitational force Example for conservative force
Frictional force Example for Non Conservative force



KE of a dropped body is gradually Increased to approach the ground PE of a dropped body is gradually decreased to approach the ground
PE = mgh (height is decreased for freely falling body)

Level – II

1) D	2) B	3) A	4) B	5) C	6) D	7) C
8) B	9) C	10) B	11) C	12) C	13) A	14) C
15) B	16) B	17) A	18) C	19) D	20) B	21) B
22) A	23) D	24) A	25) A,B	26) D	27) B	

28) a-q; b-r; c-s; d-p

1. $\frac{1}{2}\mathrm{mv}^2 = \mathrm{K.E}$
$\frac{1}{2}(50)v^2 = 625$
$v^2 = 25$

2. $v = 54$ km/h
$= 54 \times \frac{5}{18} = 15$ m/s
$\mathrm{m} = 40$ kg
$\mathrm{K.E} = \frac{1}{2}\mathrm{mv}^2 = \frac{1}{2} \times 40 \times (15)^2$

3. $\text{K.E}\ \alpha v^2$
$\text{K}_1 : \text{K}_2 = 1 : 4$
4. $\text{K} \cdot \text{E} = \text{mas} = \text{FS} = 10 \times 1 = 10\ \text{J}$
5. $\text{K.E} = \frac{1}{2}\ \text{m}(\vec{v} \cdot \vec{v})$
$\bar{v} = 2\hat{\imath} + 5\hat{\jmath}$
$v = \sqrt{4 + 25} = \sqrt{29}$
$\text{K.E} = \frac{1}{2} \times \frac{120}{1000} \times 29 = 1.745\hat{\jmath}$
6. $\text{E} = \frac{1}{2} \times \text{mv}^2$
$\frac{1}{} \times 10 \times 2$
7. $\text{K} \cdot \text{E} = \frac{1}{2}\text{mv}^2 = \text{FS} = 10 \times 1 = 10\ \text{J}$
8. $\text{w} = \text{FS} = 4 \times 2\ \text{J}$
9. $v = \frac{8}{10}; K.E = \frac{1}{} \times 3$
10. Let $\text{V}_1 = 6\ \text{V}$ Here, $\text{K.E} = \frac{1}{2}\text{mV}^2$

$$\Rightarrow \text{KE}_1 = \frac{1}{2}\text{mV}_1^2 = \frac{1}{2}\ \text{m}(6\ \text{V})^2 = 36 \times \frac{1}{2}\text{mV}^2$$

$$\Rightarrow \text{KE}_1 = 36\text{KE}$$

13. $K \cdot E_{\text{mean}} = P \cdot E_{\text{extreme}} = mgl(1 - \cos\theta)$
14. $KE = \frac{1}{2}mv^2 = \frac{1}{2}mg^2t^2\ (\because v = gt)$
15. $KE = \frac{1}{2}mv^2 = \frac{1}{2}(\rho V)v^2; \frac{K \cdot E}{V} = \frac{1}{2}\rho v^2$
16. $\text{K} = \text{F}x \Rightarrow \text{f} = \frac{\text{K}}{x}$
17. $\text{K} \cdot \text{E} = \text{FS}$
$\frac{1}{2}\text{mv}^2 = \text{FS}$
$\text{v}^2 \propto \text{s}$
$\frac{\text{s}_1}{\text{S}_2} = \frac{\text{v}_1^2}{\text{v}_2^2}$
$a = \frac{(300)^2}{2 \times 2 \times 10^{-2}} = \frac{90 \times 10^6}{4}$
$\Rightarrow \text{a} = \frac{\text{v}^2}{2 \times \text{s}} = \frac{9 \times 10^6}{A_2} = \frac{v^2}{2 \times 18_9 \times 10^{-2}}$
$v^2 = 9 \times 9 \times 10^4 \Rightarrow v = 9 \times 10^2 = 900\ \text{m/s}$
18. $\frac{\text{v}-\text{u}}{\text{t}} = \text{a} \Rightarrow \frac{\text{v}}{\text{t}} = \text{g} \Rightarrow \text{v} = \text{gt}$
$\text{t} = 3,\ \text{m} = 3\ \text{kg}, \text{u} = 0$
$\text{a} = 9.8\ \text{m/s}^2$
$v = 3 \times 9.8$ then
$\text{K.E} = \frac{1}{2}\text{mv}^2 = \frac{1}{2} \times 3 \times (3 \times 9.8)^2 = 1296.5\ \text{J}$
19. $\frac{\text{K} \cdot \text{E}}{\text{t}} = \frac{1}{2}\ \text{A}\left(\frac{l}{\text{t}}\right)\rho v^2 = \frac{1}{2}\ \text{A}\rho v^3$
20. $\frac{\text{K.E}}{4} = \text{F}x$
$\text{K.E} = 4\text{Fx} \Rightarrow \text{F.S} = 4\text{Fx} \Rightarrow \text{S} = 4\text{x}$

21. $(\text{K.E})_{\text{heavier}} = \frac{1}{2}(\text{K.E})_{\text{lighter}}$
$\frac{1}{2}\text{mv}_\text{h}^2 = \frac{1}{2}\left(\frac{\text{m}}{2}\right)\text{v}_\text{L}^2 \times \frac{1}{2} \Rightarrow \text{v}_\text{h} = \frac{\text{v}_\text{L}}{2}$

$$\frac{1}{2}\,m(v_h+2)^2=\frac{1}{2}\left(\frac{m}{2}\right)v_L^2$$

$$\Rightarrow (v_h+2)^2=4v_h^2\times\frac{1}{2}$$

$$v_h^2+4v_h=2v_h^2\Rightarrow v_h^2-4v_h-4=0$$

$$v_h=\frac{4\pm\sqrt{16+16}}{2}$$

$$v_h=2\pm\sqrt{8}$$

$v_h=2+2\sqrt{2}, v_h=2-2\sqrt{2}$

$v_h=2(1+\sqrt{2})\text{m/s}$

Speed never negative

22. $F_a-F_0=ma\Rightarrow 25-10=5a$

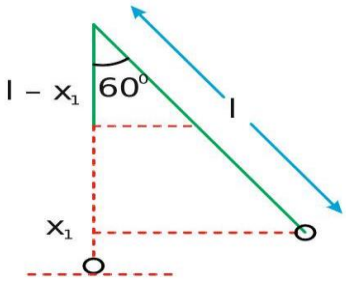
$a=3\ \text{m/s}^2$

$f_{ms}=\frac{2}{10}\times 5\times 10=10\ \text{N}$

$v^2=2as=2\times 3\times 10=60$

$\text{K.E}=\frac{1}{2}mv^2=\frac{1}{2}\times 5\times 60=150\ \text{J}$

23.



$\cos\theta=\frac{l-x_1}{l}\Rightarrow x_1=l(1-\cos 60)$

$x_2=l(1-\cos 30°)$

$x_1-x_2=l-l\cos 60°-l+l\cos 30°$

$=l(-\cos 60°+\cos 30°)$

$x=1\left(\frac{\sqrt{3}}{2}-\frac{1}{2}\right)=\frac{0.732}{2}$

$\text{K.E}=mgx=\frac{2}{10}\times 10\times\frac{0.732}{2}=0.732\ \text{J}$

24. $\Rightarrow v=\left(\frac{F}{m}\right)t=\frac{5}{10}\times 10=5\ \text{m/s}$

$\Rightarrow \text{K.E}=\frac{1}{2}mv^2=\frac{1}{2}\times 10\times 25=125\ \text{J}$

25. a) $K\cdot E=\frac{1}{2}mv^2=\frac{1}{2}\times 1\times 100=50\ \text{J}$

b) $\Delta\text{K.E}=0$

26. Conceptual

27. Conceptual

28. Conceptual

WORKSHEET – 5
Level – I

1) A	2) A	3) A	4) B	5) A	6) B	7) A
8) A	9) A	10) C				

1. The product of force and velocity is called power
   $P = FV$
2. Power is inversely proportional to time (if work = constant)
   For a given work time is less, power is more

Person	A	B
Time	15 s	20 s
Power	P_A	P_B

$\therefore P_A > P_B$

6. $\frac{w}{t} = \frac{180}{60} = 3w$
7. $w = pt$
   $\Rightarrow W = 50 \times 2 \times 3600$
   $= 360 \times 10^3$ J $= 3.6 \times 10^5$ J
8. $P = FV = 3.6 \times 10^5 \times 100 \times \frac{5}{18} = 10^8$ W
9. $\frac{P_1}{P_2} = \frac{t_2}{t_1} = \frac{15}{10} = \frac{3}{2}$
10. $t = \frac{w}{p} = \frac{3000}{60} = 50$ s

Level – II

1) D	2) A	3) C	4) B	5) C	6) A	7) B
8) D	9) B	10) C	11) B	12) D	13) C	14) D
15) B	16) A	17) C	18) A	19) B	20) D	21) A
22) C	23) B	24) A	25) A	26) –1		

1. Force = 200 N, Velocity = 20 ms^{-1}
   Power = F.V.
   $= 200 \times 20 = 4000$ Watts
2. $P = FV = 4500 \times 2 = 9000$ W $= 9$ kW
3. $P = FV = F\left(\frac{F}{m}\right)t \Rightarrow P \propto t; v = at = \left(\frac{F}{m}\right)t$
4. $P = FV = F\left(\frac{F}{m}\right)t \Rightarrow P \propto t$
5. Instantaneous Power = Ins. Force × Ins. velocity
   $= (5\vec{\imath} - 3\vec{\jmath} + 6\vec{k})(10\vec{\imath} + 10\vec{\jmath} + 20\vec{k})$
   $= (5 \times 10)\hat{\imath} - (3 \times 10)\hat{\jmath} + (6 \times 20)\hat{k}$
   $= 50\hat{\imath} - 30\hat{\jmath} + 120\hat{k}$
   $= (50 - 30 + 120)(\hat{\imath} - \hat{\jmath} + \hat{k})$
   $= 140$ W
   Power is the dot product of force and velocity.
6. $M_1 = 80$ kg, $t_1 = 15$ s
   $\Rightarrow M_2 = 80$ kg, $t_2 = 20$ S
   Power of $m_1 \Rightarrow P_1 = \frac{w_1}{t_1}$
   Power of $m^2 \Rightarrow P_2 = \frac{w_2}{t_2}$ (here $w_1 = w_2$)
   $$\frac{P_1}{P_2} = \frac{t_2}{t_1} = \frac{4}{3}$$
7. $P = FV = (mg)V = 110 \times 9.8 \times 2 = 2156$ W

8. P = F.V

$$\therefore F = \frac{P}{V} = \frac{7500}{20} = 375N$$

9. $1\text{KWH} = \frac{12\times746\times8\times10}{1000}$

$= 716.16\text{KWH} \times 0.50\text{Ps}$

$=$ Rs. 358.08Ps

10. P = constant

$w \propto t \Rightarrow FS \propto t \Rightarrow S = \frac{1}{2}at^2$

$S = \frac{1}{2}\left(\frac{F}{m}\right)t^2$

$\Rightarrow \frac{2Sm}{t^2} \times S \propto t \Rightarrow s^2 \propto t^2$

$s \propto t^{3/2}$ (or) $W \propto t$

$\frac{1}{2}mv^2 \propto t$

$\frac{1}{2}m\left(\frac{s}{t}\right)^2 \propto t$

$\frac{1}{2}\ m\frac{s^2}{t^2} \propto t$

$s^2 \propto t^3 \Rightarrow s \propto t^{3/2}$

11. $P = \frac{w}{t} = \frac{mgh}{t} = \frac{2000\times9.8\times12}{60} = 400 \times 9.8w$

$$= \frac{400 \times 9.8}{746} = 5.25\text{hp} \simeq 5.3\text{hp}$$

12. $\Rightarrow P = \frac{W}{t} = \frac{\frac{1}{2}mv^2}{t} = \frac{n\left(\frac{1}{2}mv^2\right)}{t}$

$$= \frac{240 \times \frac{1}{2} \times 5 \times 10^{-2} \times (500)^2}{60}$$

$$= \frac{}{120 \times 5 \times 250000 \times 10^{-2}}$$

$= 10 \times 2500 = 25000$ W

13. $P = \frac{W}{t} = mg\sin\theta \times v$

$= 1500 \times 10 \times \frac{1}{10} \times 5 = 7.5$ kW

$P = \frac{W}{t}$

14. $2 = \frac{n(2.4)}{60}$

$n = \frac{2 \times 60}{2.4} = \frac{60}{1.2} = 50$

$\therefore n = 50$

15. $P = \frac{\frac{1}{2}mv^2 - \frac{1}{2}mu^2}{t}$

$P = \frac{w}{t} = \frac{\Delta KE}{t}$

$$\frac{\frac{1}{2}(2.05 \times 10^6) \times (25)^2 - \frac{1}{2}(2.05 \times 10^6) \times (5)^2}{5 \times 60}$$

$$= \frac{\frac{1}{2}(2.05 \times 10^6)(625 - 25)}{5 \times 60}$$

$= \frac{(2.05 \times 10^6)(600)}{600}$

$= 2.05 \times 10^6 = 2.05\text{MW}$

16. CONCEPTUAL

17. CONCEPTUAL

18. CONCEPTUAL

19. $P = \frac{mgh + \frac{1}{2}mv^2}{t}$

$= \frac{800\left(10 \times 10 + \frac{1}{2} \times 400\right)}{60} = \frac{800 \times 300}{60} = 4000\ \text{W}$

20. $F = \frac{P}{V} = \frac{460}{S}t = \frac{460}{200} \times 117.56 = 270\ \text{N}$

21. $P = \frac{W}{t} = \frac{mgh}{t} = \frac{1000 \times 9.8 \times 6 \times 3.5}{6}$

$= 34300\ \text{W} = 34.3\ \text{kW}$

22. $P = \frac{W}{t} = \frac{1}{2} \times \frac{mgh}{t}$

$= \frac{1}{2} \times \frac{1800 \times 10 \times 9.8 \times 50}{3600}$

$= 25 \times 9.8 \times 50$

(n) $100 = 25 \times 9.8 \times 50$

$\Rightarrow n = 12.5 \times 9.8 = 122.5\ \text{W}$

23. $P = FV = mav$

$= m\left(\frac{v_1}{t_1}\right)(at) = m\left(\frac{v_1}{t_1}\right)\left(\frac{v_1}{t_1}\right)t = \frac{mv_1^2}{t_1^2}t$

24. $P = \frac{\frac{1}{2}mv^2}{t} \Rightarrow \frac{1}{2} \times \frac{2000}{4} \times v^2 \Rightarrow v = 2\ \text{m/s}$

25. $P = \frac{\frac{1}{2}mv^2}{t} \Rightarrow \frac{1}{2} = \frac{1}{2} \times \frac{2}{10} \times \frac{v^2}{5} \Rightarrow v = 5\ \text{m/s}$

26. $P = \frac{W}{t} \Rightarrow \frac{148}{10^n} = \frac{7500 \times 1050 \times 9.8 \times 1.63}{24 \times 3600}$

$\frac{148}{10^n} = 1456 \Rightarrow 10^n = \frac{148}{1456} = \frac{1}{9.8} \Rightarrow n = -1$

WORKSHEET – 6

Level – I

1. B
2. B
3. A
4. B
5. B

1. According to Work - Energy Theorem statement work done by all the forces (External &Internal) on a system is equal to change in kinetic energy.
2. Work - Energy Theorem can also be applied to a system under the action of variable forces, conservative, as well as Non - Conservative force. Hence work done on a particle (or) system is equal to the change in Kinetic Energy.
3. CONCEPTUAL
4. A stone falls from a certain height on sand if Resistance offered to its motion is less if it penetrates more. Option *A* is wrong:
   Resistance offered to its motion is more penetrates is less it depends on the extent penetration.
5. $m = 0.5\ \text{kg}, v = 2\ \text{m/s}, s = 2\ \text{m}$
   $W = FS = mas$
   $v^2 - u^2 = 2as$

$$a = \frac{v^2 - u^2}{2\,s}$$
$$a = \frac{-u^2}{2\,s}$$
$$a = \frac{(-2)^2}{2s}$$
$$= \frac{-4}{2 \times 2}$$
$$= \frac{-4}{4} = -1$$
$a = -1$(retardation)
$a = 1$
$W = 0.5 \times 1 \times 2$
$$= \frac{5}{10} \times 2 = \frac{5}{5} = 1$$
$W = 1$ J

Level – II

1) B	2) C	3) C	4) A	5) B	6) B	7) C
8) D	9) A	10) B	11) A	12) B	13) D	14) C
15) C	16) B	17) B	18) B	19) A,B,D	20) B	21) A

1. Mass of scooter = 86 kg
Mass of scooter + Petrol
= 86 + 64 = 150 kg
According to work energy theorem
$$W_{all} = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$
$$K.E = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$
$u = 36$ km/h
$= {}^{2}36 \times \frac{5}{18} = 10$ m/s v = 18 km/h
$$= 18 \times \frac{5}{18} = 5 \text{ m/s}$$
$$= \frac{1}{2}(150)(10^2 - 5^2) = 75(100 - 25)$$
$= 75 \times 75 = 5625$ J

2. $F = 10$ N,
$u = 5$ m/s, m = 4 kg
$v = 5$ m/s
According to work energy theorem
$$W = \Delta K \cdot E = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$
$$F.S = \frac{1}{2}m(v^2 - u^2)$$
$$\therefore S = \frac{\frac{1}{2}\,m(v^2 - u^2)}{F} = \frac{\frac{1}{2}(4)(10^2 - 5^2)}{10}$$

3. According to work energy theorem
$w = \Delta K.E$

$w = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

$w_1 = m(v^2 - u^2)$

$= \frac{1}{2}m(20^2 - 10^2)$

$= \frac{1}{2}m(400 - 100)$

$w_1 = \frac{1}{2}\ m(300) \rightarrow (1)$

$w_2 = \frac{1}{2}\ m(10^2 - 0^2)$

$w_2 = \frac{1}{2}\ m(100) \rightarrow (2)$

$\frac{(1)}{(2)}$

$\frac{w_1}{w_2} = \frac{1/2\ m(300)}{1/2\ m(100)} = \frac{300}{100}$

$\therefore w_1 = \frac{300 \times w_2}{100}$

$\therefore w_1 = 3w_2$

$\therefore$ 3 times of w_2

4. $mgh - w_{air} = \frac{1}{2}mv^2$

$5 \times 10 \times 20 - w_{air} = \frac{1}{2} \times 5 \times (10)^2$

$1000 - w_{air} = 250$

$w_{air} = 1000 - 250 = 750$ J

5. $FS = \frac{1}{2}\ m(v^2 - u^2)$

$F \times d_1 = \frac{1}{2} \times \frac{1}{100}(200 \times 200 - 180 \times 180)$

$F = \frac{1}{200}(400 - 324)$

$= \frac{76}{2} = 38$ N

(or)

$-Fs = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

$-F = \frac{1}{2} \times 10^{-2}[180^2 - 240^2]$

$F = \frac{1}{200}[40000 - 32400]$

$F = \frac{76}{2} = 38$ N

6. $mgh - w_{air} = \Delta KE$

$1 \times 10 \times 200 - w_{air} = \frac{1}{2} \times 1 \times (50)^2$

7. Mass of ship $= 3 \times 10^7$ kg

$u = 0$

$s = 3$ m

$F = 5 \times 10^4$ N

$W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

$Fs = \frac{1}{2}mv^2 - 0$

$mv^2 = 2Fs$

$v^2 = \frac{2Fs}{m}$

$V = \sqrt{\frac{2Fs}{m}} = \sqrt{\frac{2 \times 5 \times 10^4 \times 3}{3 \times 10^7}}$

$= \sqrt{10^{5-7}} = \sqrt{10^{-2}} = 10^{-1}$

$= 0.1$ m/s

8. $V^2 = 25 + 4\ S$

$v^2 = u^2 + 2as$

$u = 5$ m/s $\Rightarrow 2a = 4 \Rightarrow a = 2$ m/s^2

$v = u + at = 5 + 2(2) = 9$ m/s

$s = ut + \frac{1}{2}at^2$

$s = 14$ m

$W = FS = mas$

$\Rightarrow 56$ J

9, $w = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

F. $S = \frac{1}{2}m(v^2 - u^2)$

Case $(i) \Rightarrow \frac{1}{2}\ m(20^2 - 10^2) = w_1$

$w_1 = \frac{1}{2}\ m(400 - 100)$

$w_1 = \frac{300\ m}{2}$

Case $(ii) \Rightarrow \frac{1}{2}\ m(15^2 - 5^2) = w_2$

$w_2 = \frac{1}{2}\ m(225 - 25)$

$w_2 = \frac{200\ m}{2}$

$\frac{w_1}{w_2} = — \times — = \frac{300}{200}$

$\frac{w_1}{w_2} = \frac{3}{2}$

$w_1 : w_2 = 3 : 2\ w_1 = 1\frac{1}{2}w_2$

$= 1\frac{1}{2}$ times

10. $W = \Delta KE \Rightarrow KE_i = F_{ret}\ S$

11. $W = FS = \triangle KE$

12. $W = FS = \Delta KE$

13. $W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

14 Work done by the net force is equal to change in KE. Work-Energy Theorem is applicable not only for a single particle but also for a system of particles.

15. $w_f = \frac{1}{2}\ m(v^2 - u^2) = \frac{1}{2} \times 70 \times ((5)^2 - (2.5)^2)$

16 Work done will be same

$$\frac{1}{2}\,m((200)^2-(180)^2)=\frac{1}{2}\,m((180)^2-v^2)$$

$$=100(400-324)=100\times100-v^2$$

$$v^2=1000-7600$$

$$v^2=2400\Rightarrow v=49\ m/s$$

Work done independent of initial speed. Same mass, same plank. So w is same for both cases.

17,18 $\cdot\ V=\sqrt{2gh}$

$$=\sqrt{2\times10\times3.2}$$

$$=\sqrt{6.4\times10}=\sqrt{64}=8\ m/sec$$

From law of conservation of energy.

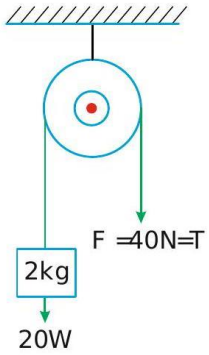
Loss in P.E = gain in K.E

$$mgh=\frac{1}{2}mv^2$$

$$\therefore v=\sqrt{2gh}$$

$$=\sqrt{2\times10\times3.2}$$

$$=\sqrt{6.4\times10}=\sqrt{64}=8\ m/sec$$



19.

$$40-20=2a\Rightarrow a=10\ m/s^2$$

$$K.E=\frac{1}{2}mv^2=40$$

$$\frac{1}{2}\times2\times v^2=40\Rightarrow v^2=40$$

$$v^2-2as\Rightarrow40=2\times10\times S\Rightarrow S=2\ m$$

$$w_g=-(mg)S=-(20)2=-40\ J$$

$$w_T=(40)^2=80\ J$$

20. Mass of a bullet m = 10 g

$$=\frac{10}{1000}=0.01\ kg$$

$$w_{air}\ \text{resistance}=\frac{1}{2}mv^2-\frac{1}{2}mu^2$$

$$=\frac{1}{2}0.01\times(10{,}00{,}000-2{,}50{,}000)$$

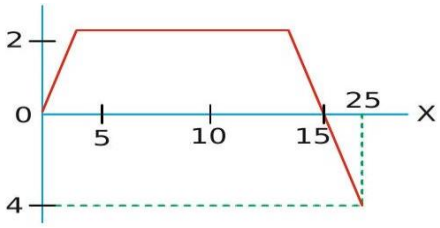
$$=\frac{1}{2}0.01\times7{,}50{,}000$$

$$=3{,}75{,}000\times0.01$$

$$=3750\ J$$

F (Newton)

21. a)



maximum K.E = Area under the graph above x - axis

$= \frac{1}{2} \times 5 \times 2 + 5 \times 2 + \frac{1}{2} \times 3 \times 2$

$= 5 + 10 + 5 = 20$ J at 15 m b) K. E at 25 m

$K.E = 20 + \left(-\frac{1}{2} \times 10 \times 4\right) = 0$

Speed is zero at $x = 25$ m

c) K. E at 5 m

$K = \frac{1}{2} \times 2 \times 5 = 5$ J

d) K. E at 10 m

$K = \frac{1}{2} \times 2 \times 5 + 5 \times 2 = 15$ J

WORKSHEET – 7

Level – I

1) C 2) B 3) D 4) D 5) 5

1. The negative of the work done by the conservation internal forces on a system equals the change in potential energy.
   Option A, B, D are wrong:
   $W_C = -(v_f - v_i)$
   $W_C \neq \Delta K$
   $W_C \neq E_2 - E_1$

2. $W_c + W_{nc} + W_{ext} \neq K_f - K_i$
   According to work-Energy theorem work done by all the forces (conservative, Non-conservative, external) on a system is equals the change in KE.
   Option A, C, D are wrong
   $W_c + W_{nc} + W_{ext} \neq v_f - V_i$
   $W_c + W_{nc} + W_{ext} \neq \Delta E$

3. According to law of conservation of energy
   $W_C = -(v_f - v_i)$
   $W_{nc} + W_{ext} = E_f - E_i$
   According to work - energy theorem
   $W_c + W_{nc} + W_{ext} = \Delta K$
   Option D is wrong
   $W_c + W_{nc} + W_{ext} \neq U_f - U_i$

4. $W_c + W_{nc} + W_{ext} = K_f - K_i$
   If the Kinetic energy of a body is increasing work done by conservative force may be positive may be zero
   Work done by Non-conservative forces may be zero

5. Speed is a scalar quantity it does not depends upon direction of the stone.

6. $\frac{490}{2} = mgh \Rightarrow \frac{490}{2} = 9.8 \times 2 \times h$
$\Rightarrow h = 12.25$ m

Level – II

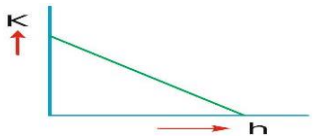
1) A	2) D	3) B	4) B	5) C	6) A	7) A
8) A	9) D	10) B	11) C	12) B	13) B	14) A
15) B	16) C	17) C	18) D	19) C	20) D	21) B,C
22) A	23) a-p; b-r; c-s; d-q		24) a-q; b-q; c-p; d-q,s		25) C	26) A
27) A	28) D	29) A	30) A	31) B		
32) a-q,r,s; b-s; c-q,r,s; d-p,r,s			33) a-q; b-r; c-s,p; d-p			

2. m = 50 kg,
$l = 15$ m $\times \frac{l}{3} = 5$ m
K.E. =?
Loss in P.E = gain in K.E
$mhgh = \frac{1}{2}mv^2$
$v = \sqrt{2gh}$
$= \sqrt{2 \times 10 \times 5} = \sqrt{100} = 10$ m/s
$K.E = \frac{1}{2}mv^2$
$= \frac{1}{} \times 50 \times {}^{50}100 = 2500$ J

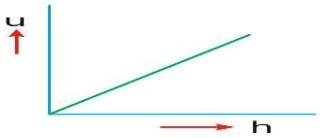
3. Sol; $Mgx = \frac{1}{2}kx^2 \Rightarrow x = \frac{2Mg}{k}$
$EPE = \frac{1}{2}kx^2$

4. $\frac{1}{2}m\left(\frac{v^2}{4} - v^2\right) = -\frac{1}{2}kx^2$
$m\left(\frac{3v^2}{4}\right) = kx^2 \Rightarrow k = \frac{3mvv^2}{4x^2}$

5. $\frac{1}{2}kx^2 = mg(h + x)$
$\frac{1}{2}k(0.1)^2 = 1 \times 10(50 \times 10^{-2})$
$k = \frac{10}{0.01} = 1000$ N/m

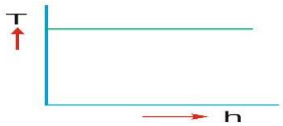
6. 

Height increases K.E decreases

7. 

Height increases P.E increases

8.



T.E remains constant

$\frac{1}{2}\ m\left(\frac{v^2}{2} - v^2\right) = mgh$

9. $\Rightarrow \frac{v^2}{4} = gh \Rightarrow h = \frac{v^2}{4\,g}$

10. $mgh = \frac{1}{2}mv^2\ gh = \frac{1}{2}v^2$

$\therefore v = \sqrt{2gh} = \sqrt{2 \times 10 \times 0.5}$

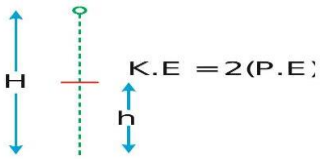
$\sqrt{5 \times 2} = \sqrt{10}$ m/sec

11. $v = \sqrt{u^2 + 2gh}$

$= \sqrt{2500 + 2 \times 10 \times 40} = \sqrt{3300}$

$= 57.49$ m/s $\simeq 58$ m/s

12.



$\frac{1}{2}mv^2 = 2PE$

$\frac{1}{2}\ m \times 2\ g(H - h) = 2mgh$

$H - h = 2\ h \Rightarrow H = 3\ h$

$h = H/3$

$v = \sqrt{2g(H - h)} = \sqrt{2\ g\left(\frac{2H}{3}\right)} = 2\sqrt{\frac{gH}{3}}$

13. KE = PE changes

$\frac{1}{2}mv^2 = mgh_2 - mgh_1$

$\frac{1}{2}v^2 = gh_2 - gh_1$

$v^2 = 2\ g(\ h_2 - h_1)$

$v = \sqrt{2\ g(\ h_2 - h_1)}$

$= \sqrt{2 \times 10(2 - 0.75)} = \sqrt{20 \times 1.25} =$

$= \sqrt{25} = 5$ m/sec

14. $v = \sqrt{2\ g(\ h_2 - h_1)}$

$= \sqrt{2 \times 10 \times 0.1} = \sqrt{2} = 1.414$ m/sec.

15. $mg(h + x) = \frac{1}{2}kx^2$

$2 \times 9.8(0.4 + x) = \frac{1}{2} \times 1960 \times x^2$

$100x^2 - 2x - 0.8 = 0$

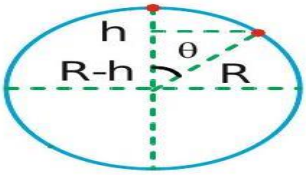
$100x^2 - 10x + 8x - 0.8 = 0$

$10x(10x - 1) + 0.8(10x - 1) = 0$

$10x - 1 = 0\ 10x + 0.8 = 0$

$10x = 1 \Rightarrow x = \frac{1}{10} = 0.1$ m

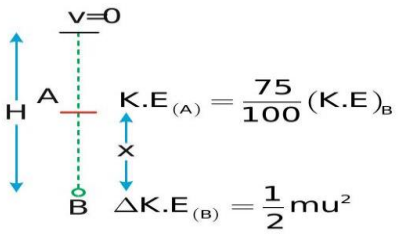
16.



$\cos\theta = \dfrac{R-h}{R}; h = R(1-\cos\theta)$

$\Delta P.E = P.E_f - P.E_i = mgh = mgR(1-\cos\theta)$

17.



$u^2 = 2gh$

$(K.E)_A = \dfrac{3}{4}(K.E)_B$

$\dfrac{1}{2}m(u^2 - 2gx) = \dfrac{3}{4}\left(\dfrac{1}{2}mu^2\right)$

$4u^2 - 8gx = 3u^2 \Rightarrow u^2 = 8gx$

$2gh = 8gx$

$x = h/4$

18. $\frac{1}{2}mv^2 = \frac{9}{10}(mgh)$

$v^2 = \dfrac{9}{8} \times 10 \times 2$

$v = 6$ m/s

19. Mechanical energy is of two types
(1)PE
(2) KE
A body at Rest cannot have Kinetic Energy but it can have potential energy. For a freely falling body PE is decreases & KE is Increases. So Mechanical energy of a freely falling body remains constant

20. The total energy of a system is conserved. According to law of conservation of energy but here given to Total energy of a system is always conserved irrespective of whether
external forces act on the system is wrong. If external force acting on the system, the total momentum and energy will increased.

21. $W_c + W_{nc} + W_{ext} = K_f - K_i$

22. $E = mc^2$

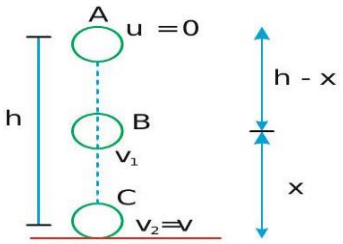
$C = 3 \times 10^8$ m/s

$m = 10^{-3}$ kg

$= 10^{-3} \times 9 \times 10^{16}$

$= 9 \times 10^{13}$ J

23. A body is allowed to fall from a height ' h ' from the ground as shown in figure



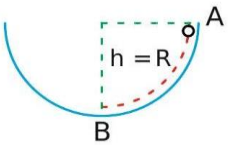
P) PE = KE
Potential energy at point 'B'
PE = mg x
KE at point ' B ' is

$$\mathrm{KE} = \frac{1}{2}\mathrm{mv}_1^2$$
$$\mathrm{v}^2 - \mathrm{u}^2 = 2\mathrm{as}$$
$$\mathrm{v}_1^2 - \mathrm{o}^2 = 2g(\mathrm{h} - x)$$
$$\mathrm{v}_1^2 = 2\,\mathrm{g}(\mathrm{h} - x)$$
$$\therefore \mathrm{KE} = \mathrm{mg}(\mathrm{h} - x)$$
$$\therefore \mathrm{PE} = \mathrm{KE} \Rightarrow \mathrm{mg}x = \mathrm{mg}(\mathrm{h} - x)$$
$$x = \mathrm{h} - x$$
$$x = \mathrm{h}/2$$

At height h/2PE = KE
Q) PE = 2KE ⇒ mg$x$ =
mgg x = 2mgg(h − x)
x = 2 h − 2x
3x = 2h
$x = \frac{2h}{3}$
At height PE = 2KE
R) PE = 2KE ⇒ mg(h − x) = 2mg$x$
$h - x = 2x$
$h = 3x$
x = h/3
At height h/3KE = 2PE
S) Sum PE and KE is constant at any point on the path followed by freely falling body.

$$\mathrm{v} = \sqrt{\mathrm{u}^2 + 2\mathrm{gh}}$$
$$= \sqrt{2500 + 2 \times 10 \times 40}$$
$$= \sqrt{3300} = 57.49 \text{ m/s} \simeq 58 \text{ m/s}$$

26.



$\mathrm{KE_B} = \mathrm{PE_A}$
$\frac{1}{2}\mathrm{mv}^2 = \mathrm{mgh}$

27. $\mathrm{v} = \sqrt{2\mathrm{gh}} = \sqrt{2 \times 9.8 \times \frac{1}{10}} = 1.4$ m/s



$KE_A = KE_B + KE_B$

$\frac{1}{2}mu^2 = mgL + \frac{1}{2}mv^2$

$v = \sqrt{u^2 - 2gL}$

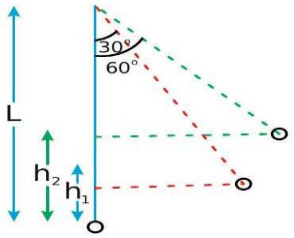
Magnitude of change in velocity

$= \sqrt{v^2 + u^2 + 2uv\cos 90°}$

$= \sqrt{(u^2 - 2gL) + u^2}$

$\Delta v = \sqrt{2(u^2 - gL)}$

29.



$\cos 60° = \frac{L - h_2}{L} = \frac{1}{2} \Rightarrow 2\,L - 2\,h_2 = L \Rightarrow L = 2\,h_2$

$h_2 = L/2$

$\cos 30° = \frac{L - h_1}{L} = \frac{\sqrt{3}}{2} \Rightarrow 2\,L - 2\,h_1 = \sqrt{3}\,L$

$\Rightarrow 2\,h_1 = (2 - \sqrt{3})L \; h_1 = \frac{(2-\sqrt{3})L}{2}$

$K.E = mg(h_2 - h_1)$

$= mg\left(\frac{1}{2} - \frac{1}{2}(2 - \sqrt{3})\right)L$

$= mg\frac{L}{2}(1 - 2 + \sqrt{3})$

$= mg\frac{L}{2}(\sqrt{3} - 1)$

$= \frac{2}{10} \times 9.8 \times \frac{1}{2}(0.782)$

$= 0.7174\ J = 7.174 \times 10^6 erg$

$P.E = mgh = \frac{2}{10} \times 9.8 \times \left(\frac{2-1.732}{2}\right) \times 1$

$= 0.26264\ J$

$= 2.626 \times 10^6 erg$

30. $v = \sqrt{2gh} = \sqrt{2 \times 10 \times 80} = \sqrt{1600} = 40\ m/s$

31. $mgh = \frac{1}{2}kx^2$

$\Rightarrow \frac{4}{100} \times 9.8 \times 4.9 = \frac{1}{2} \times 100 \times x^2$

$x^2 = \frac{1}{100} \times \frac{1}{100} \times 9.8 \times 9.8$

$x = \frac{9.8}{100} = 9.8\ cm$

6. GRAVITATION

WORKSHEET – 1

Level – I

1) C	2) D	3) A	4) A	5) B	6) B	7) C
8) A	9) C	10) A				

Level - II

1) D	2) D	3) B	4) C	5) B	6) A	7) D
8) C	9) A	10) B	11) B	12) C	13) A	14) C
15) A	16) C	17) C	18) C	19) a-t;b-q;c-p;d-r	20)A,B,C	
21) A,B,C	22) C	23) C	24) A	25) a-p;b-q;c-s;d-q		

1. $F = G\frac{mM}{d^2}$

   $F' = \frac{G(3m)M}{d^2} + \frac{GmM}{d^2}$

   $= \frac{\mathrm{GMm}}{\mathrm{d}^2}[3+1] = 4$ F = xf

   x = 4

2. The gravitational force is maximum when the mass is divided into equal parts

3. $\mathrm{F} = (6.67 \times 10^{-11})\frac{(1\times1)}{10^{-4}}$

   $\mathrm{F} = 6.67 \times 10^{-7}$ N

4. $\mathrm{F} = \mathrm{G}\frac{\mathrm{m_1\ m_2}}{\mathrm{r}^2}$

   $\mathrm{F} = \mathrm{G}\cdot\frac{\mathrm{m}\cdot\mathrm{m}}{\mathrm{d}^2}$

6. $\mathrm{F}\alpha\frac{1}{\mathrm{r}^2}$

   Force decreases by 4 times

   $\mathrm{F} = \frac{1}{2}$ N

7. Intensity of gravitational field inside the follow spherical shell is zero

8. $\vec{\mathrm{E}}_{\mathrm{net}} = \overrightarrow{\mathrm{E}_1} + \overrightarrow{\mathrm{E}_2}$

   $= \frac{-G(4)}{(2)^2} + \frac{G(8)}{(4)^2} = -G + \frac{G}{2} = -\frac{G}{2}$

   $\left|\vec{\mathrm{E}}_{\mathrm{net}}\right| = \frac{\mathrm{G}}{2}$

9. The vectors $\overrightarrow{\mathrm{E}_1}, \overrightarrow{\mathrm{E}_2}, \overrightarrow{\mathrm{E}_3}$

Form a closed polygon (triangle)

Hence resultant gravitational field $\vec{E} = 0$

10. $\frac{G\times10^3}{x^2} = \frac{G\times10^5}{(x-2)^2}$

$$\frac{1}{x^2} = \frac{10^2}{(x-2)^2}$$

$$(x-2)^2 = 100 \cdot x^2$$

$$9x = -2$$

$$x = \frac{-2}{9}$$

$$x - 2 = -10x$$

$$11x = 2$$

$$x = \frac{2}{11}m$$

11. $G = 6.67 \times 10^{-11}$ N

$F = 6.67 \times 10^{-7}$ N

$$F = G\frac{(m)(4m)}{(10^{-1})^2}$$

$$6.67 \times 10^{-7} = \frac{6.67 \times 10^{-11} \times 4\ m^2}{10^{-2}}$$

$m = 5$ kg

$4\ m = 20$ kg

12. $F_1 = G\frac{(3)(1)}{1^2} = 3G$

$$F_2 = G\frac{(4)(1)}{1^2} = 4G$$

$$\left|\vec{F}_{net}\right| = \sqrt{F_1^2 + F_2^2 + 2\ F_1\ F_2 \cos\theta}$$

$\theta = 90°, \cos 90° = 0$

$$F_{net} = \sqrt{(3G)^2 + (4G)^2} = 5G$$

13. $F \propto \frac{1}{r^2}$

The distance is increased by 100% i.e., distance is doubled.

The gravitational force becomes $\frac{F}{4}$

14. $F = G\frac{m_1 m_2}{r^2}$

$$F_A = \frac{G(20)(30)}{(1)^2} = 600G$$

$$F_B = \frac{G(40)(60)}{\left(\frac{1}{2}\right)^2} = 9600G$$

$$F_c = \frac{G(30)(50)}{(2)^2} = 375G$$

$$F_D = \frac{G(10)(40)}{(2.5)^2} = 64G$$

F_B, F_A, F_c, F_D

15. $r_1 = r\ r_2 = r + \frac{r}{4} = \frac{5r}{4}$

$$\mathrm{F}\alpha\frac{1}{\mathrm{r}^2}$$

$$\frac{F_2}{F_1} = \frac{r_1^2}{r_2^2} = \frac{r^2}{25\frac{r^2}{16}}$$

$$\frac{\mathrm{F}_2}{\mathrm{F}_1} = \frac{16}{25} = -1$$

$$\frac{\mathrm{F}_2 - \mathrm{F}_1}{\mathrm{F}_1} = \frac{-9}{25} \times 100 = \frac{-9}{25} \times 100 = -36\%$$

Gravitational force decreases by 36%

16. $\mathrm{F}_1 = \mathrm{F},$

$r_1 = r$

$$= \frac{64\ \mathrm{F}}{100} \Rightarrow r_2 = r + 3$$

$$\frac{\mathrm{F}_2}{\mathrm{F}_1} = \frac{\mathrm{r}_1^2}{\mathrm{r}_2^2}$$

$$\frac{64\frac{F}{100}}{F} = \frac{r^2}{(r+3)^2} = \frac{64}{100} = \left(\frac{r}{r+3}\right)^2$$

$$\frac{r}{r+3} = \pm\frac{8}{10} = \frac{r}{r+3} = \frac{8}{10}$$

$10r = 8(r + 3) \Rightarrow 10r = 8r + 24$

$2\mathrm{r} = 24 = 12\ \mathrm{m}$

17. The gravitational field anywhere inside the spherical shell is zero.

Hence, Force exerted is zero

22. $F_{\text{net}} = F_1 + F_2$

$$= \frac{G(1)(2)}{(1)^2} + \frac{G(1)(3)}{(2)^2} = \frac{11}{4}G$$

$$= \frac{11 \times 6.67 \times 10^{-11}}{4} = 18.34 \times 10^{-11}$$

23. $F_{\text{net}} = |F_1 - F_2|$

$$= \left|\frac{G(1)(2)}{(1)^2} - \frac{G(2)(3)}{(1)^2}\right| = G|2-6| = 4G$$

$$= 4 \times 6.67 \times 10^{-11} = 26.68 \times 10^{-11} \text{ N}$$

24. $F_{\text{net}} = F_1 + F_2$

$$= \frac{G(1)(3)}{(2)^2} + \frac{G(2)(3)}{(1)^2} = \frac{27}{4}G$$

$$= 45.02 \times 10^{-11} \text{ N}$$

25. (a) $F_{\text{net}} = \frac{G(m)(m)}{\left(\frac{\sqrt{3}a}{2}\right)^2} = \frac{4Gm^2}{3a^2}$

(b) $F_{\text{net}} = 0$

(c) $F_{\text{net}} = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta} = \sqrt{3}F = \frac{\sqrt{3}Gm^2}{a^2}$

(d) $F_{\text{net}} = 0$

WORKSHEET – 2

Level – I

1) C 2) A 3) B 4) C 5) A 6) A 7) A
8) B

Level – II

1) C	2) A	3) D	4) A	5) C	6) A	7) B
8) B	9) C	10) C	11) D	12) D	13) B	14) C
15) D	16) a-r;b-q;c-p;d-s		17) A,C	18) A,B,C,D	19) A	20) A
21) B	22) B	23) C	24) C			

1. $\frac{g'}{g} = \frac{R^2}{(R+h)^2}$

$$g' = \frac{g}{2} \Rightarrow \frac{1}{2} = \frac{R^2}{(R+h)^2}$$

$$\frac{1}{2} = \frac{R^2}{(R+h)^2}$$

$$(R+h)^2 = 2R^2$$

$$R + h = \sqrt{2}R$$

$$\text{R} + \text{h} = (\sqrt{2} - 1)\text{R} \Rightarrow \text{h} = 0.414\text{R}$$

2. $g_d = g\left(1 - \frac{d}{R}\right)$

$$g_d = \frac{g}{4}$$

$$\frac{g}{4} = g\left(1 - \frac{d}{R}\right)$$

$$\frac{d}{R} = \frac{3}{4} = \frac{3R}{4}$$

$$= \frac{3}{4}(6400) = 3 \times 1600 = 4800 \text{ km}$$

3. $g' = \frac{64}{100} \times g$

$$\frac{g'}{g} = \frac{64}{100} = \frac{R^2}{(R+h)^2}$$

$$\frac{\mathrm{R}}{\mathrm{R}+\mathrm{h}} = \frac{8}{10}$$

$$10R = 8R + 8h$$

$$2\mathrm{R} = 8\ \mathrm{h}$$

$$\mathrm{h} = \frac{2\mathrm{R}}{8} = \frac{\mathrm{R}}{4} = \frac{6400}{4} = 1600 \text{ km}$$

4. $g' = g - R\omega^2\cos^2\lambda$

$$g^1 - g = R\omega^2\cos^2\lambda$$

$$\frac{3}{4}\mathrm{R}\omega^2 = \mathrm{R}\omega^2\cos^2\lambda$$

$$\frac{\sqrt{3}}{2} = \cos\lambda \Rightarrow \lambda = 30°$$

5. $\mathrm{g}_\lambda = \mathrm{g} - \mathrm{R}\mathrm{w}^2\cos^2\lambda$

$$\mathrm{g}_\lambda = 0,\ \lambda = 45°$$

$$g = (R\cos^2\lambda)w^2$$

$$g = \frac{R}{2}w^2$$

$$w = \sqrt{\frac{2g}{R}}$$

6. $w = mg$

$$\frac{g'}{g} = \frac{R^2}{(R+R)^2} = \frac{R^2}{4R^2}$$

$$g' = \frac{g}{4}$$

$$w' = mg' = \frac{mg}{4} = \frac{w}{4}$$

7. $g_d = g\left(1 - \frac{d}{R}\right)$

$$d = \frac{R}{2}$$

$$g_d = \frac{g}{2}$$

$$mg_d = \frac{mg}{2} = \frac{100}{2} = 50\ N$$

8. $g = \frac{GM}{R^2}$

$$\frac{g}{5} = \frac{GM_1}{\left(\frac{R}{4}\right)^2} - - - -($$

dividing (1) & (2)

$$\frac{M}{m_1} = \frac{80}{1}$$

9. $g_d = g\left(1 - \frac{d}{R}\right)$

$$\frac{g}{n} = g\left(1 - \frac{d}{R}\right)$$

$$\frac{1}{n} = 1 - \frac{d}{R} \Rightarrow \frac{d}{R} = 1 - \frac{1}{n}$$

$$d = \left(\frac{n-1}{n}\right)R$$

10. $g = \frac{GM}{R^2} = \frac{GM_p}{\left(\frac{Dp}{2}\right)^2}$

$$g = \frac{4GM}{p}$$

11. $\frac{g'}{g} = \frac{R^2}{(R+h)^2}$

$$g' = \frac{g}{16}$$

$$\frac{1}{16} = \frac{R^2}{(R+h)^2}$$

$$\frac{R}{R+h} = \frac{1}{4}$$

$4R = R + h \Rightarrow h = 3R$

12. $1 - \frac{2h}{R} = 1 - \frac{d}{R}$

$d = 2\ h$

13. $\frac{g_h}{g} = \left(\frac{R}{R+h}\right)^2$

$$\frac{g_h}{g} = \left(\frac{R}{R\left(1 + \frac{1}{10}\right)}\right)^2$$

$$\frac{g_h}{g} = \frac{100}{121}$$

$$g_h = \frac{100\ g}{121}$$

14. $\frac{g_h}{g} = \left(\frac{1}{1+\frac{1}{30}}\right)^2 \Rightarrow g_h = \frac{900g}{961}$

15. $g_h = g\left(\frac{1}{1+\frac{1}{20}}\right)^2 \Rightarrow g_h = \left(\frac{400}{441}\right)g$

16. $\frac{\Delta g}{g} = \frac{d}{R} \times 100\%$

$= \frac{3200}{6400} = \frac{1}{2}\% = 0.5\%$

20. $\frac{g^1}{g}\% = \frac{d}{R} \times 100$

$0.9 = \frac{d}{R}$

$d = 0.1 \times 6400$

21. $\frac{g^1}{g^{11}} = \frac{\left(\frac{R}{5}\right)}{\left(\frac{R}{10}\right)} = \frac{2}{1}$

22. $\frac{3}{4} = \frac{d}{R}$

$d = \frac{3R}{4}$

23. $\frac{g^1}{g} = \left(\frac{R}{R+h}\right)^2$

$\frac{1}{4} = \left(\frac{R}{R+h}\right)^2 \Rightarrow \frac{1}{2} = \frac{R}{R+h}$

$R + h = 2R \Rightarrow h = 2R - R$

$h = R = 6400km$

24. $\frac{g^1}{g} = \frac{1}{4} \times 6400 = 16\%$

25. $g = \frac{GM}{R^2} \Rightarrow g = \frac{G\frac{4}{3}\pi R^3 \rho}{R^2}$

$g = \frac{4}{3}G\pi R\rho \Rightarrow G = \frac{3g}{4\pi R\rho}$

26. $g = \frac{4}{3}G\pi Rd, g\alpha R \Rightarrow \frac{Rm}{Re} = \frac{Rm}{Re} = \frac{1}{6}$

$\frac{R_m}{R_e} = \frac{g_m}{g_e} \times \frac{\rho_e}{\rho_m} = \frac{1}{6} \times \frac{5}{3} = \frac{5}{18}$

WORKSHEET – 3

Level – I

1) B 2) B 3) B 4) A 5) D 6) C

Level – II

1) 2	2) 1	3) 3	4) 3	5) 1	6) 4	7) 2
8) 2	9) 2	10) 1	11) 4	12) 2,3	13)1	14) 4
15)2	16) a-s;b-r; c-q; d-p		17) 1	18) 2	19)3	20) 1,3
21) a-p; b-q; c-r; d-s	22) a-q;b-s;c-p;d-s			23) 3	24) 2	25) 1

1. $V_{\text{centre}} = \frac{3}{2}V_{\text{surface}}$

$$W = m(V_f - V_i)$$
$$= m(V_{\text{surface}} - V_{\text{centre}})$$
$$= m\left(V_{\text{surface}} - \frac{3}{2}V_{\text{surface}}\right)$$
$$= +m\left(\frac{-V_{\text{surface}}}{2}\right)\left(\because V_{\text{surface}} = \frac{-GM}{R}\right)$$
$$= \frac{-1}{2}m\left(\frac{-GM}{R}\right) = \frac{-1}{2}m\left(\frac{-GM}{R}\right)$$
$$= \frac{1}{2}mgR \left(\because g = \frac{GM}{R^2}\right)$$

2. $U_i = \frac{-GMm}{R}$

$$U_f = \frac{-GMm}{3R}$$

$$\mathrm{W} = \mathrm{U_f} - \mathrm{U_i} = \frac{-\mathrm{GMm}}{3\mathrm{R}} - \left(\frac{-\mathrm{GMm}}{\mathrm{R}}\right)$$
$$= \frac{-GMm + 3GMm}{3R} = \frac{2GMm}{3R}$$

$$V_{\text{surface}} = \frac{-GM}{R}$$

4. $U_i = \frac{-\mathrm{GMm}}{R}$

$$U_f = \frac{-\mathrm{GMm}}{(R+3R)} = \frac{-\mathrm{GMm}}{4R}$$

$$\Delta U = U_f - U_i = \frac{-\mathrm{GMm}}{4R} - \left(\frac{-\mathrm{GMm}}{R}\right)$$
$$= \frac{-\mathrm{GMm}}{4R} + \frac{\mathrm{GMm}}{R}$$
$$= \frac{-\mathrm{GMm} + 4\mathrm{GMm}}{4R}$$
$$= \frac{3\mathrm{GMm}}{4R}$$
$$= \frac{3}{4}mgR \left(\because g = \frac{GM}{R^2}\right)$$

5. $U_i = \frac{-\mathrm{GMm}}{R}$

$$U_f = \frac{-\mathrm{GMm}}{(R+nR)}$$
$$\Delta = U_f - U_i$$
$$= \frac{-\mathrm{GMm}}{R+nR} - \frac{-\mathrm{GMm}}{R}$$
$$= \frac{-R\cdot \mathrm{GMm} + \mathrm{R}\cdot \mathrm{GMm} + \mathrm{Nr}\cdot \mathrm{GMm}}{R(R+nR)}$$
$$= \frac{nRGMm}{R(R+nR)}$$
$$= \left(\frac{n}{n+1}\right)\frac{GMm}{R}$$
$$= \left(\frac{n}{n+1}\right)mgR \ \left(\because g = \frac{GM}{R^2}\right)$$

6. $\Delta T \cdot E = \Delta KE + \Delta U$

Work done $= \Delta T.\mathrm{E}$

$= -10$ J

$\Delta K.E = 4J$

$-10J = +4K + \Delta U$

$AU = -14J$ per unit mass $= \frac{-14}{2}$

$= -7$ J/kg

7. Potential Energy of particle $\mathrm{m_4}$

$$U_i = U_1 + U_2 + U_3$$
$$= \frac{-Gm_1m_4}{a} - \frac{Gm_3m_4}{a} - \frac{Gm_2m_4}{\sqrt{2}a}$$

$\mathrm{m_1} = \mathrm{m_2} = \mathrm{m_3} = \mathrm{m_4} = 1$ kg

a = 1 m

$$u_i = -G\left(1+1+\frac{1}{\sqrt{2}}\right)$$

$$\mathrm{u_i} = -\mathrm{G}\left(\frac{2\sqrt{2}+1}{\sqrt{2}}\right)$$

$\mathrm{u_f} = 0$

$$\mathrm{w} = \mathrm{u_f} - \mathrm{u_i} = \mathrm{G}\left(\frac{2\sqrt{2}+1}{\sqrt{2}}\right)$$

8. $U = -G\left[\frac{m_1m_2}{a} + \frac{m_1m_3}{a} + \frac{m_2m_3}{a}\right]$

$m_1 = 1$ kg, $m_2 = 2$ kg, $m_3 = 3$ kg

$$U = -G\left[\frac{(1)(2)}{20\times10^{-2}} + \frac{(1)(3)}{20\times10^{-2}} + \frac{(2)(3)}{20\times10^{-2}}\right]$$

$$U = -G\left[\frac{2+3+6}{20\times10^{-2}}\right]$$

$U = -G[55] = -55GJ$

9. $F_{net} = \sqrt{3}\ F$

$F = \frac{Gm^2}{r^2}$

$F_{net} = \frac{\sqrt{3}Gm^2}{r^2}$

10. All the vectors of gravitational field intensity forms a closed polygon. (Triangle).

Hence the resultant is Zero

11. The Gravitational potential due to any one particle at the centroid is $V = \frac{Gm}{\left(\frac{r}{\sqrt{3}}\right)} = \frac{\sqrt{3}Gm}{r}$

The gravitational potential due to all three particles = 3 V $V = \frac{3\sqrt{3}Gm}{r}$

13. $\Delta U = -\int F dr$

$E = \frac{GMx}{(a^2 + x^2)} 3/2$

14. $x = 0 \Rightarrow E = 0$

15. x << a

$E = \frac{GMx}{(a^2 + x^2)^{3/2}} = \frac{GMx}{(a^2)^{3/2}}$

$E = \frac{GM}{a^3} x$

$a\alpha - x$

17. $V = \frac{-GM}{R} = \frac{-6.67\times10^{-11}\times5.98\times10^{24}}{6.37\times10^{6}}$

$= 6.26 \times 10^7$ J/kg

18. $U = mV = 0.1 \times -6.26 \times 10^7 = 6.26 \times 10^6$ J

19. $U = \frac{-GM}{(R+h)}$

$h = R$

$U = \frac{-GM}{(2R)} = \frac{-6.26 \times 10^7}{2} = 3.13 \times 10^7$ J

20. $v_i = -\frac{GM}{R} = v$

$$v_f = -\frac{GM}{(R+h)} = -\frac{GM}{2R}$$

$$\Delta v = v_f - v_i = -\frac{GM}{2R} + \frac{GM}{R}$$

$$= \frac{GM}{2R}$$

$$= -\frac{v}{2}$$

Similarly $\Delta U = -\frac{u}{2}$

23. $U = \frac{-GMm}{(R+h)}$

$$\mathrm{U_i} = \frac{-\mathrm{GMm}}{\mathrm{R}}$$

$$U_f = \frac{-GMm}{(R+3R)} = \frac{-GMm}{(4R)}$$

$$\mathrm{W} = \Delta\mathrm{U} = \mathrm{U_f} - \mathrm{U_i} = \frac{-\mathrm{GMm}}{(4\mathrm{R})} - \left(\frac{-\mathrm{GMm}}{\mathrm{R}}\right)$$

$$\mathrm{W} = \frac{3\mathrm{GMm}}{4\mathrm{R}} = \frac{3}{4}\mathrm{mgR}$$

24. $\mathrm{U_i} = \frac{-\mathrm{GMm}}{\mathrm{R}}$

$$U_f = \frac{-GMm}{\left(R+\frac{R}{3}\right)} = \frac{-3GMm}{(4R)}$$

$$W \Rightarrow \Delta U = U_f - U_i = \frac{-3GMm}{(4R)} - \left(\frac{-GMm}{R}\right)$$

$$\mathrm{W} = \frac{\mathrm{GMm}}{4\mathrm{R}} = \frac{1}{4}\mathrm{mgR}$$

25. $U_\mathrm{i} = \frac{-\mathrm{GMm}}{\mathrm{R}}$

$$U_f = \frac{-GMm}{(R+nR)} = \frac{-GMm}{R(n+1)}$$

$$\mathrm{W} = \Delta\mathrm{U} = \mathrm{U_f} - \mathrm{U_i} = \frac{-\mathrm{GMm}}{\mathrm{R(n+l)}} - \left(\frac{-\mathrm{GMm}}{\mathrm{R}}\right)$$

$$W = \frac{-GMm}{R}\left(\frac{1}{n+1} - 1\right)$$

$$\mathrm{W} = \frac{\mathrm{GMmn}}{\mathrm{R(n+1)}}$$

$$\mathrm{W} = \frac{\mathrm{mgRn}}{(\mathrm{n+1})}$$

WORKSHEET – 4

Level – I

1) A 2) A 3) A 4) B 5) C 6) A

Level – II

1) D	2) B	3) A	4) B	5) C	6) A	7) C
8) B	9) B	10) D	11) A	12) D	13) B	14) A
15) B	16) A	17) A	18) C	19) C	20) A,B,C	
21) A,B,C,D	22) a-s; b-r; c-p; d-q					

1. Given $F = \frac{Gm_e m_s}{r^2} = m_e r\omega^2$

$$\omega^2 = \frac{Gm_s}{r^3}$$

$$\left(\frac{2\pi}{T}\right)^2 = \frac{Gm_s}{r^3}$$

$$T^2 = \frac{4\pi^2 r^3}{Gm_s}$$

$$m_s = \frac{4\pi^2 r^3}{T^2 \cdot G}$$

Time period = 1 year

$= 365 \times 24 \times 60 \times 60$sec

$$m_s = \frac{4 \times (3.14)^2 \times (1.5 \times 10^{11})}{6.67 \times 10^{-11} \times (365 \times 24 \times 60 \times 60)^2}$$

$= 2 \times 10^{30}$ kg

2. $v_e = \sqrt{\frac{2GM}{R}}$

$r = 1.74 \times 10^6$ m.

$m = 7.4 \times 10^{22}$

$G = 6.63 \times 10^{-11} \text{Nm}^2\ \text{kg}^{-2}$

$$V_e = \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 7.4 \times 10^{22}}{1.74 \times 10^6}} = 2.4 \text{ km/s}$$

3. The escape velocity of a body depends upon mass so escape velocity $V_e = \sqrt{2gR}$. So it is independent of mass. Therefore it depends on m^0

4. $V_e = \sqrt{\frac{2GM}{R}}$

$$\omega = \frac{V}{R} = \frac{1}{R}\sqrt{\frac{2GM}{R}} = \sqrt{\frac{2GM}{R^3}}$$

5. $v_e = \sqrt{\frac{2GM}{R}}$

$$V_e = \sqrt{\frac{2G}{R} \cdot \frac{4}{3}\pi R^3 \cdot \rho}$$

$$v_e = \sqrt{\frac{8G\pi}{3} R^2 \rho}$$

$$v_e \propto R\sqrt{\rho}$$

$$\frac{v_e}{v_p} = \frac{R\sqrt{\rho}}{(2R)\sqrt{2\rho}}$$

$$v_e : v_p = 1 : 2\sqrt{2}$$

6. $v_e \propto R\sqrt{\rho}$

$$\frac{v_p}{v_e} = \frac{(2R)\sqrt{\rho}}{(R)\sqrt{\rho}}$$

$$v_p = 2v_e$$

$$v_p = 2 \times 11 = 22 \text{ km/s}$$

7. $v_0 = \sqrt{g(R + h)}$

$$v_0 = \sqrt{9.8(6.38 \times 10^6 + 0.25 \times 10^6)}$$

$$v_0 = 8.05 \text{ km/s}$$

8. $v_e \propto \sqrt{\frac{M}{R}}$

$$\frac{v_{moon}}{v_{earth}} = \sqrt{\frac{M}{R} \times \frac{4R}{81M}}$$

$$\frac{v_{moon}}{V_{earth}} = \frac{2}{9}$$

$$V_{moon} = \frac{2}{9} \times 11.2 = 2.48 \text{ km/s}$$

9. $M_p = 6M_e, R_p = 3R_e$

$$v = \sqrt{\frac{Gm}{R}}$$

$$v_e = \sqrt{\frac{Gm_e}{R_e}}$$

$$v_p = \sqrt{\frac{Gm_p}{R_P}}$$

$$\frac{v_p}{v_e} = \sqrt{\frac{m_p}{R_p} \times \frac{R_e}{m_e}} = \sqrt{\frac{6 \times m_e}{3R_c} \times \frac{R_e}{m_e}}$$

$v_p = \sqrt{2}v$

10. The condition for the space ship to escape out of earth gravitational pull

T.E = 0

$$-\frac{GMm}{R} + E = 0$$

$$E = \frac{GMm}{R}$$

$$= \frac{6.67 \times 10^{-11} \times 6 \times 10^{24} \times 10^3}{64 \times 10^5}$$

$E = 6.4 \times 10^{10}$ J

11. $v_e = \sqrt{2gR}$

12. $v = \sqrt{2\ g \cdot R} = \sqrt{2 \times 9.8 \times 6.4 \times 10^6}$

$= 11.2$ km/s

13. R = 1,737.1 km

$$v = \sqrt{2 \times \frac{9.8}{6} \times 1737.1 \times 10^3} = 2.38 \text{ km/s}$$

14. $v_0 = \sqrt{\frac{GM}{(R+h)}}$

15. $v_0^2 = \frac{GM}{(R+h)}$

But given that $v_0 = \frac{1}{2}v_e$

$$v_0^2 = \frac{1}{4}v_e^2$$

$$\frac{GM}{(R+h)} = \frac{1}{4}\left(\frac{2GM}{R}\right)$$

$2R + 2\ h = 4R$

$h = R = 6400$ km

16. When the satellite is stopped, its kinetic energy is zero. When it falls freely on the earth, its potential energy decreases and converts into ki netic energy.

$$\frac{-GMm}{2R} - \left(\frac{-GMm}{R}\right) = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{GM}{R}} = \sqrt{gR} = \sqrt{9.8 \times 6.4 \times 10^6}$$

$v = 7920$ m/s $\Rightarrow = 7.92$ km/s

17. $TE_f = \frac{-GMm}{2(3R)} = \frac{-GMm}{6R}$

Energy required $= TE_f - TE$;

$$= \frac{-GMm}{6R} + \frac{-GMm}{4R}$$

$$= \frac{-2GMm + 3GMm}{12R}$$

$$= \frac{GMm}{12R}$$

18. $U = 2(T.E)$

$U = 2E_0$

19. $TE_1 = -\frac{GMm}{R} + E$

$$TE_2 = -\frac{GMm}{2(R + 2R)}$$

$$= -\frac{GMm}{6R}$$

$TE_1 = TE_2$

$$-\frac{GMm}{R} + E = -\frac{GMm}{6R}$$

21. P. $E_{initial} = 0$

K. $E_{initial} = 0$

T.E $E_{initial} = 0$

since energy is conserved total energy is always zero (option- 1 is correct)

Since there is no external torque acting, the net angular momentum is zero (OPTION-2 is correct)

Since energy is conserved, the decrease in P.E of the system = increase in K.E of the system

$$0 - \left(\frac{-G(m)(4\ m)}{r}\right) = K.E$$

$K.E = \frac{4Gm^2}{r}$ (OPTION-3 is correct)

Gain in K.E $= \frac{4Gm^2}{r}$

$$\frac{1}{2}\mu v_r^2 = \frac{4Gm^2}{r}$$

Where $\mu = \frac{m_1 m_2}{m_1 + m_2}$ called reduced mass

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{m \cdot 4m}{m + 4m} = \frac{4m}{5}$$

$$\frac{1}{2}\left(\frac{4\ m}{5}\right)v_r^2 = \frac{4Gm^2}{r}$$

$$v_r = \sqrt{\frac{10GM}{r}}$$

WORKSHEET – 5

Level – I

1) B	2) C	3) A	4) B	5) B

Level – II

1) D	2) D	3) C	4) B	5) C	6) B	7) A
8) B	9) D	10) A	11) D	12) D	13) A	14) B
15) C	16) A	17) 5	18) D			

1. From kepler's 3rd Law
$T^2 \alpha r^3$
$\frac{T1^2}{T_2^2} = \frac{r_1^3}{r_2^3}$
$\frac{T^2}{T_2^2} = \frac{R^3}{(3R)^3}$
$T_2^2 = 27T^2$
$T_2 = 3\sqrt{3}T$

2. $\frac{r_1}{r_2} = \frac{1}{2}$
from keplers 3rd law $T^2 \alpha r^3$
$$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3} = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$
$$\frac{T_1}{T_2} = \frac{1}{2\sqrt{2}}$$

3. $T^2 \propto R^3$
$\left(\frac{T_n}{T_s}\right)^2 = \left(\frac{10^{13}}{10^{12}}\right)^3$
$\frac{T_n}{T_s} = \sqrt{1000} = 10\sqrt{10}$

4. $T^2 \propto R^3$
$\left(\frac{T}{1\ day}\right)^2 = \left(\frac{R}{R}\right)^3$
$T^2 = \frac{1}{64} = \frac{1}{8}$

5. From kepler's 3rd law $T^2 \propto r^3$
$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3}$
$\frac{(24)^2}{T_2^2} = \frac{r^3}{(2r)^3}$
$T_2^2 = (24)^2(8) \Rightarrow T_2 = 48\sqrt{2}$

6. $F = \frac{Gm_1m_2}{r^2}$
$F = mr\omega^2$

$F \propto R^{\frac{-5}{2}}$

$F = KR^{\frac{-5}{2}}$

$\mathrm{ma} = \mathrm{KRR}^{\frac{-5}{2}}$

$a = \frac{K}{m} R^{\frac{-5}{2}} \left(\because \omega = \frac{2\pi}{T}, T = \frac{2\pi}{\omega}\right)$

$a \propto R^{\frac{-5}{2}}$

$a = m\omega^2$

$\omega^2 R \propto R^{\frac{-5}{2}}$

$\omega^2 \propto \mathrm{R}^{\frac{-7}{2}}$

$\omega \propto \mathrm{R}^{\frac{-7}{4}}$

$\frac{2\pi}{T} \propto R^{\frac{-7}{4}} \Rightarrow \frac{T}{2\pi} \propto R^{\frac{7}{4}} \Rightarrow T \propto R^{\frac{7}{4}}$

$\mathrm{T}^2 \propto \mathrm{R}^{\frac{7}{2}}$

7. $\mathrm{md_1v_1} = \mathrm{md_2v_2}$

$v_2 = \frac{d_1 v_1}{v_2}$

8. $\mathrm{T}^2 \propto \mathrm{R}^3$

$\frac{1}{n^2} = R^3$

$\mathrm{n}^2\mathrm{R}^3 = \text{constant}$

$\mathrm{n}_1^2\ \mathrm{d}_1^3 = \mathrm{n}_2^2\ \mathrm{d}_2^3$

9. Time period is independent of mass of the satellite, but depends on radius.

$\mathrm{T}^2 \propto \mathrm{R}^3$

$\left(\frac{\mathrm{T}_1}{\mathrm{T}_2}\right)^2 = \left(\frac{\mathrm{r}}{2\mathrm{r}}\right)^3$

$\frac{\mathrm{T}_1}{\mathrm{T}_2} = \frac{1}{2\sqrt{2}}$

10. $\mathrm{mvr} = \text{constant}$

$\mathrm{vr} = \text{constant}$

$v \times 8 \times 10^{12} = 1.6 \times 10^{12} \times 60 = 12\ \mathrm{m/s}$

11. From keplers 3^3 d law$_T$ $^2\alpha r^3$

$T^2 = ky^3$

Toy 3/2

12. $\mathrm{L} = \mathrm{mvr}$

$\mathrm{L} = m\sqrt{\frac{GM}{r}} r$

$L = m\sqrt{GMr}$

$L \propto \sqrt{r}$

$\frac{L_2}{L_1} = \sqrt{\frac{r_2}{r_1}}$

$\frac{L_2}{L} = \sqrt{\frac{16r}{r}} \Rightarrow L_2 = 4L$

13. $\mathrm{T.E} = -\frac{\mathrm{GMm}}{2\mathrm{a}}$

$\mathrm{T.E} = \mathrm{K.E} + \mathrm{P.E}$

$$-\frac{GMm}{2a} = \frac{1}{2}Mv^2 - \frac{GMm}{r}$$

$$-\frac{Gm}{2a} = \frac{v^2}{2} - \frac{Gm}{r}$$

$$\mathrm{Gm}\left[\frac{2}{\mathrm{r}} - \frac{1}{\mathrm{a}}\right] = \mathrm{v}^2$$

$$v = \sqrt{Gm\left[\frac{2}{r} - \frac{1}{a}\right]}$$

17. $\mathrm{T}^2 \alpha \mathrm{R}^3$

$$\frac{\mathrm{T}_1^2}{\mathrm{T}_2^2} = \frac{\mathrm{R}_1^3}{\mathrm{R}_2{}^3}$$

$$\frac{9}{\mathrm{T}_2^2} = \frac{\mathrm{R}^3}{(4\mathrm{R})^3}$$

$\mathrm{T}_2 = 5 \times$ times

$\mathrm{T}_2^2 = 64 \times 9$

$$25x^2 = 64 \times 9 \Rightarrow x^2 = \frac{64 \times 9}{25}$$

$$x = \sqrt{\frac{34 \times 9}{25}} = \frac{8 \times 3}{5} = 4.8 \simeq 5$$

7. HEAT

WORKSHEET – 1

Level – I

1) B	2) D	3) B	4) C	5) D	6) A	7) B
8) B	9) C	10) B	11) C	12) A		

Level – II

1) B	2) A	3) B	4) C	5) A	6) D	7) A
8) A	9) D	10) B	11) C	12) B	13) B	
14) A,B,C	15) C	16) B	17) A	18) C	19) a-q,b-s,c-p,d-r	
20) C	21) A,B	22) B	23) C	24) A		

1. $l_1 = 2\text{ m}, l_2 = 2.0024\text{ m},$
$t_1 = 27°C, t_2 = 77°C$
$\alpha = \frac{l_2 - l_1}{l_1(t_2 - t_1)} = \frac{2.0024 - 2}{2(77 - 27)} = \frac{0.0024}{2 \times 50} = 24 \times 10^{-6}/\,^0C$

2. $l_1 = 1\text{ cm}, l_2 = 1.02\text{ cm}$
$\Delta t = 100°C$
$\alpha = \frac{l_2 - l_1}{l_1(t_2 - t_1)} = \frac{1.02 - 1}{1 \times 100} = \frac{0.02}{100} = 2 \times 10^{-4}/\,^0C$
$l_1 = 2\text{ m}$
$t_1 = 10°C, t_2 = 30°C$
$t_2 - t_1 = 20°C$

3. $\alpha = \frac{l_2 - l_1}{l_1(\Delta T)}$
$l_2 - l_1 = 40 \times 10^{-4}$
$l_2 - l_1 = 0.004\text{ cm}$
Original length $= l_1 - (l_2 - l_1)$
$= 2 - 0.004$
$= 1.996\text{ m}$

4. $\alpha = \frac{l_2 - l_1}{l_1(t_2 - t_1)}$
$l_2 = 101\text{ cm}, l_1 = 100\text{ cm}$
$t_2 - t_1 = 100\ °C$
$\alpha = \frac{101 - 100}{100 \times 100} = \frac{1}{10000}$
$\Rightarrow 1 \times 10^{-4}/K$

5. $\alpha = \frac{l_2 - l_1}{l_1 \Delta t}$
$= \frac{200.24 - 200}{200 \times 60}$
$= \frac{0.24}{200 \times 60}$
$= 2 \times 10^{-5}/\,^0C$

6. $l_1 = 50\text{ cm}, l_2 = 50.12\text{ cm}$
$t_1 = 12^0c$
$\alpha = 12 \times 10^{-6}$
$\alpha = \frac{l_2 - l_1}{l_1 \Delta t}$

$$12 \times 10^{-6} = \frac{50.12 - 50}{50(t_2 - t_1)}$$
$$50 \times 12 \times 10^{-6} = \frac{0.12}{t_2 - 12}$$
$$600 \times 10^{-6} = \frac{12}{100(t_2 - 12)}$$
$$60000 \times 10^{-6} = \frac{12}{t_2 - 12}$$
$$\frac{6}{100} = \frac{12}{t_2 - 12}$$
$$t_2 - 12 = \frac{12 \times 100}{6}$$
$$t_2 = 200 + 12$$
$$= 212^{\circ}C$$

7. $A_1 = 1.2mH^2,\ A_2 - A_1 = \frac{2.4\times10^{-4}}{1.2\times(100-0)}$
$t_1 = 0^{\circ}C, t_2 = 100^{\circ}C$
$$\beta = \frac{A_2 - A_1}{A_1(t_2 - t_1)} = \frac{2.4 \times 10^{-4}}{1.2 \times (100 - 0)}$$
$$= \frac{2.4}{120} \times 10^{-4}$$
$$= \frac{2.4}{1.2} \times 10^{-6}$$
$$= 2 \times 10^{-6}/\ ^0C$$

8. $\gamma = 57 \times 10^{-6}/\ ^0C$
$v_1 = 150cc$
$v_2 - v_1 = ?$
$t_2 = 100^{\circ}C, t_{1=0}$
$$\gamma = \frac{v_2 - v_1}{v_1(t_2 - t_1)}$$
$$57 \times 10^{-6} = \frac{v_2 - 150}{150 \times 100}$$
$$0.855 = v_2 - 150$$
$$v_2 = 150.86cc$$

9. $I_2 - I_1 = 2.7\ mm = 2.7 \times 10^{-3}\ m$
$$\alpha = \frac{I_2 - I_1}{I_1(t_2 - t_1)}$$
$$90 \times 10^{-6} = \frac{27 \times 10^{-3}}{0.5 \times (t_2 - 0)}$$
$$t_2 = 60^{\circ}C$$

10. $\alpha = \frac{I_2 - I_1}{l_1(t_2 - t_1)}$
$$\alpha\alpha \frac{1}{1}$$
$$\frac{I_1}{I_2} = \frac{\alpha_2}{\alpha_1}$$
$$\frac{liron}{lbrass} = \frac{\alpha\ brass}{\alpha\ iron}$$
$$= \frac{18 \times 10^{-6}}{11 \times 10^{-6}}$$

$$\frac{\text{l iron}}{\text{l brass}} = \frac{18}{11} = 18:11$$

11. $\mathrm{I}_1 = 50$ cm
$\mathrm{t}_2 - \mathrm{t}_1 = 80 - 20 = 60°\mathrm{C}$
$\alpha = 24 \times 10^{-6}/\ ^0\mathrm{C}$
$\mathrm{I}_2 - \mathrm{I}_1 = ?$
$$\alpha = \frac{\mathrm{I}_2 - \mathrm{I}_1}{\mathrm{I}_1(\mathrm{t}_2 - \mathrm{t}_1)}$$
$\mathrm{I}_2 - \mathrm{I}_1 = \alpha \mathrm{I}_1(\mathrm{t}_2 - \mathrm{t}_1)$
$= 24 \times 10^{-6} \times 50 \times 60$
$= 3000 \times 24 \times 10^{-6}$
$= 3000 \times 24 \times 10^{-6}$
$= 7.2 \times 10^{-2}$

12. $\mathrm{I}_1 = 25$ cm
$\mathrm{b}_1 = 8$ cm
$A_1 = I_1 \times b_1 = 25 \times 8 = 200\ \mathrm{cm}^2$
$t_1 = 0^0C, t_2 = 100°\mathrm{C}$
$\beta = \frac{A_2 - A_1}{A_1(t_2 - t_1)}$
$A_2 - A_1 = \beta A_1(t_2 - t_1)$
$A_2 = -A_1[1 + \beta A_1(t_2 - t_1)]$
$= 200(1 + 36 \times 10^{-6} \times 100)$
$= 200(1 + 0.0036)$
$= 200 \times 1.0036$
$A_2 = 200.72\ \mathrm{cm}^2$

13. $r_1 = 100$ cm
$r_2 = 101$ cm
$\mathrm{t}_2 - \mathrm{t}_1 = 100 - 0 = 100°\mathrm{C}$
$\mathrm{r} = ?$
$$r = \frac{v_2 - v_1}{v_1(t_2 - t_1)} = \frac{\frac{4}{3}\pi r_2^3 - \frac{4}{3}\pi r_1^3}{\frac{4}{3}\pi r_1^3(t_2 - t_1)}$$
$$= \frac{r_2^3 - r_1^3}{r_1^3(t_2 - t_1)} = \frac{(101)^3 - (100)^3}{100^3 \times 100}$$
$$= \frac{1030301 - 1000000}{10^8}$$
$= 30301 \times 10^{-8}$
$= 30301 \times 10^{-4}/\ ^0\mathrm{C}$
$\gamma = 3 \times 10^{-4}/\ ^0\mathrm{C}$

14. OPTION - 1,2, 3 (Conceptual)
15. OPTION - 3
16. $\beta : \gamma = 2:3$
$$\frac{\beta}{\gamma} = \frac{2}{3}$$
$$\Rightarrow \beta = \frac{2}{3}\gamma = \frac{2x}{3}$$
17. $\beta = 2\alpha = 22 \times 10^{-6}\mathrm{OC}^{-1}$
18. $\gamma = 3\alpha$
$= 3 \times 19 \times 10^{-6}$
$= 57 \times 10^{-6}\mathrm{OC}^{-1}$

19. $a-2, b-4, c-1, d-3$

20. OPTION -3

21. For first rod $l_2 = l_1(1+\alpha_1\Delta T)$
For second rod $l_2 = l_2(1+\alpha_2\Delta T)$
Given that change in length is independent of temperature.
$l_2 - l_2^{|} = l_1 - l_2 - - - -(1)$
$l_2 - l_2^{|} = l_1(1+\alpha_1\Delta T) - l_2(1+\alpha_2\Delta T)$
$l_2 - l_2^{|} = (l_1 - l_2) + l_1\alpha_1\Delta T - l_2\alpha_2\Delta T -$
From (1) & (2)
$l_1\alpha_1\Delta T = l_2\alpha_2\Delta T$
$l_1\alpha_1 = l_2\alpha_2$ (or) $\frac{l_1}{l_2} = \frac{\alpha_2}{\alpha_1}$

22. $\mathrm{I_{Iron}} - \mathrm{I_{brass}} = 5$ cm
$\frac{\mathrm{I_{Iron}} - 1.5}{\mathrm{I_{Iron}}} = \frac{12}{18}$
$\frac{1.5}{\mathrm{I_{Iron}}} = \frac{1-12}{18}$
$I_{\mathrm{Iron}} = 1.5 \times \frac{18}{6} = 4.5$ cm
$\mathrm{I_{brass}} = 4.5 - 1.5 = 3.0$ cm

WORKSHEET – 2

Level – I

1) A	2) D	3) D	4) C	5) D	6) B	7) A
8) C	9) C	10) C	11) B	12) C	13) D	14) C

Level – II

1) B	2) A	3) B	4) C	5) D	6) D	7) A
8) C	9) D	10) B	11) C	12) A	13) B	14) C
15) A	16) A	17) C	18) B	19) A	20) C	21) A,C
22) a-s, b-q, c-p, d-q,r		23) A	24) A,C	25) B,D	26) a-p, b-r, c-s, d-q	
27) $\gamma = 9.02 \approx 9$		28) B	29) A	30) B		

1. Given, $\gamma = X\beta$
$3\alpha = \mathrm{X}(2\alpha)$
$x = \frac{3}{2} = 1.5$

2. We know that, $\gamma = 3\alpha$ and $\beta = 2\alpha$
$\therefore \gamma = \frac{3}{2}\beta$
$\beta = \frac{2}{3}\gamma$
$\beta = \frac{2}{3} \times 0.00027$
$\beta = 0.00018/\,^0C$

3. We know that, $1^0\mathrm{C} = 1.8°\mathrm{F}$
$1^0\ \mathrm{F} = \frac{1}{1.8}\ ^0\mathrm{C}$
given , $\alpha = 5 \times 10^{-5}/\,^0\ \mathrm{F}$

$$\alpha = \frac{5 \times 10^{-5}}{\left(\frac{1}{1.8}\right)^{0}\text{C}} = 9 \times 10^{-5}/\,^{0}\text{C}$$

$\alpha = 0.00009/\,^{0}c$

4. We know that, $\ell_\theta = \ell_\theta(1 + \gamma\Delta\theta)$

$\ell_\theta = \ell_\theta\,(1 + 3\alpha\Delta\theta)$

given, $\ell_0 = 11.34$ g/cm^3

$\alpha = 28 \times 10^{-6}/\,^{0}\text{C}$

$\ell_{100} = \ell_\theta = ?$

$\Delta\theta = 100\,^{\circ}\text{C}$

$\therefore \ell_{100} = 11.34[1 + 3 \times 28 \times 10^{-6} \times 100]$

$\ell_{100} = 11.25$ g/cm^3

5. We know that, $\beta = 2\alpha$

$$\alpha = \frac{\beta}{2}$$

$$\alpha = \frac{2 \times 10^{-5}}{2}$$

$\alpha = 1 \times 10^{-5}/\,^{0}\text{C}$

6. We know that density decreases with height

$$\frac{h_1}{h_2} = \frac{\ell_2}{\ell_1} = \frac{(1 + \gamma t_1)}{(1 + \gamma t_2)} \left[\because\ \ell = \frac{\ell_0}{(1 + \gamma\Delta T)}\right]$$

$$\frac{50}{60} = \frac{1 + \gamma \times 50}{1 + \gamma \times 100}$$

$$5 + 500\gamma = 6 + 300\gamma$$

$$200\gamma = 1$$

$$\gamma = \frac{1}{200}$$

$\gamma = 0.005/\,^{0}\text{C}$

7. $\gamma > \frac{\gamma}{3}$; The liquid level rises.

8. Surface are (A) of the sheet at $0\,^{0}\text{C}$

$= 40$ cm $\times\ 8$ cm $= 320$ cm^2

Surface are $(A + \Delta A)$ of the sheet at $100\,^{\circ}\text{C}$.

Increase in Area $(\Delta A)(320.1 - 320) = 0.1$ cm^2

$$\therefore \beta = \frac{\Delta A}{A\Delta T} = \frac{0.1}{320 \times 100} = 31 \times 10^{-7}/\,^{0}\text{C}$$

$\therefore$ **Coefficient of linear expansion**

$$\alpha = \frac{\beta}{2} = \frac{31 \times 10^{-7}}{2} = 15.5 \times 10^{-7}/\,^{0}\text{C}$$

9. We know that

$$\gamma = \frac{\Delta V}{V\Delta T} = \frac{\left(\frac{\Delta V}{V}\right)}{\Delta T}$$

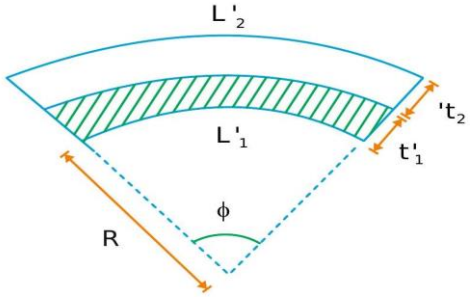
Here, $\frac{\Delta V}{V} = 0.12\% = \frac{0.12}{100}$ and $\Delta T = 20\,^{\circ}\text{C}$

$$\gamma = \frac{\left(\frac{0.12}{100}\right)}{20} = 6 \times 10^{-5}/\,^{0}\text{c}$$

$$\alpha = \frac{\gamma}{3} = \frac{6 \times 10^{-5}}{2} = 2 \times 10^{-5}/\,^{0}\text{c}$$

10. When the metallic is kept in a cold bath, the length of strip X decreases more rapidly than Y, So the bimetallic strip bends towards the left

11. at $T_0 = T_1$
Consider $t_1 = t_2 = t$
$L_1 = L_2 = L_0$
at $T_0 + \Delta T = T_2$
$\text{Angle}(\theta) = \dfrac{\text{arc}}{\text{radius}}$



$\theta = \frac{L_1^1}{t_1^1+R} = \frac{L_2^1}{t_1^1+t_2^1+R} \rightarrow (1)$

$L_1^1 = L_0(1 + \alpha_1 \Delta T)$
$L_2^1 = L_0(1 + \alpha_2 \Delta T)$
$t_1^1 = t(1 + \alpha_1 \Delta T)$
$t_2^1 = t(1 + \alpha_2 \Delta T)$

From the equations (1) and (2), by neglecting terms involving $\alpha_1, \alpha_2 \& a_1^2$, we get $R = \left[\frac{[1+\alpha,\Delta T]t}{(\alpha_2-\alpha_1)(\Delta T)}\right]$

12. Coefficient of volume expansion of glycerine,
$\gamma = 49 \times 10^{-5} / {}^0C$
Rise in temperature, $\Delta T = 30°C$
$\gamma = \dfrac{\Delta V}{\Delta T} \Rightarrow \dfrac{\Delta V}{V} = \gamma \times \Delta T$
$\dfrac{\Delta V}{V} = (49 \times 10^{-5} / {}^0C)(30°C)$
$\dfrac{\Delta V}{V} = 1.47 \times 10^{-3}$
Since mass remains constant, fractional change in density = fractional change in volume.
$= 1.47 \times 10^{-2} = 1.5 \times 10^{-2}$

13. $\rho = \frac{m}{V}$
By differentiating, we get
$\dfrac{d\rho}{dT} = \left(-\dfrac{1}{v^2}\right)\left(\dfrac{dv}{dT}\right)m$
$\dfrac{d\rho}{dT} = -\left(\dfrac{m}{v}\right)\left(\dfrac{dv/v}{dT}\right)m$
$\dfrac{d\rho}{dT} = -\rho\gamma$
$\gamma = \left(\dfrac{-1}{\rho}\right)\left(\dfrac{d\rho}{dT}\right)$

14. $L_x = L_0(1 + \alpha_x \Delta T), L_y = (1 + \alpha_y \Delta T)L_0$
$L_z = L_0(1 + \alpha_z \Delta T)$
Thus, $V = L_x L_y L_z$
$V = L_0^3(1 + \alpha_x \Delta T)(1 + \alpha_y \Delta T)(1 + \alpha_z \Delta T)$
Neglecting $\alpha_x\alpha_y, \alpha_y\alpha_z, \alpha_z\alpha_x$ and $\alpha_x\alpha_y\alpha_z$

$$V = V_0\left(1 + (\alpha_x + \alpha_y + \alpha_z)\Delta T\right)$$
$$V = V_0[1 + \gamma\Delta T]$$
$$\gamma = \alpha_x + \alpha_y + \alpha_z$$

15. $L^1 = R^1\theta = L_0(1 + \alpha_1\Delta T) \rightarrow (1)$
$L^{11} = R^{11}\theta = L_0(1 + \alpha_2\Delta T) \rightarrow (2)$
Where R^1, R^{11} are the radii of curvature of the two strips and θ is the angle subtended at the centre of the bimetallic strip.
From 1,2 equations
$L^{11} - L^1 = (R^{11} - R^1)\theta = (\alpha_2 - \alpha_1)L_0\Delta T$
$= d\theta = (\alpha_2 - \alpha_1)L_0\Delta T$
$\theta = \dfrac{(\alpha_2 - \alpha_1)L_0\Delta T}{d} \rightarrow$
If R is means Radius
$R^1 + R^{11} = 2R$
$2R\theta = 2\ L_0 + (\alpha_2 + \alpha_1)L_0\Delta T$
$R = \dfrac{2\ L_0 + (\alpha_2 + \alpha_1)L_0\Delta T}{2\theta} \rightarrow (4)$
From 3, 4 equations
$R = \left[\dfrac{2 + (\alpha_2 + \alpha_1)\Delta T}{2(\alpha_2 - \alpha_1)\Delta T}\right]d$
Neglecting $(\alpha_2 + \alpha_1)\Delta T$ in comparing to (2) equation we get

$$R = \frac{d}{(\alpha_2 - \alpha_1)\Delta T}$$

16. Percentage change in length with change in temperature = %I
$\dfrac{\Delta l}{l} \times 100 = \alpha\Delta\theta \times 100 = 1$
change in area
$\%\ A\%\ A = \dfrac{\Delta A}{A} \times 100 = \beta\Delta\theta \times 100 = 2(\alpha\Delta\theta \times 100)$
$\%\ A = 2\%$
change in volume
$\%\ V = \dfrac{\Delta V}{V} \times 100 = V\Delta\theta \times 100 = 3(\alpha\Delta\theta \times 100)$
$\%\ V = 3\%$

17. Δt just sink, gravitational force = upthrust force
$mg = F_b$
$v\rho_1\gamma = v\rho_2\gamma$
$\rho_1 = \rho_2$
$$\frac{880}{1 + 1.2 \times 10^{-3}\theta} = \frac{900}{1 + 1.5 \times 10^{-3}\theta}$$
By simplyfing, we get
$\theta = 83°C$

18. Let L, σ, h be the side of cube, density of the mercury and length of floating cube inside the mercury at 27 °C of 300 K
By principle of floatation
$m = v_{in}\sigma$
$m = hL^2\sigma \rightarrow (1)$
If side of cube, density of mercury and length of the floating side of cube inside the liquid are $L^1, \sigma^1,\ h^1$ at 100 °C of 373 K, then

$$m = v_{in}^{1}\sigma^{1}$$
$$m = h^{1}(L^{1})^{2}\sigma^{1}$$
$$\therefore hL^{2}\sigma = h^{1}(L^{1})^{2}\sigma^{1} \rightarrow (2)$$
$$\frac{h^{1}}{h} = \left(\frac{L}{L^{1}}\right)^{2} \times \frac{\sigma}{\sigma^{1}}$$

Now, $L^{1} = L(1 + \alpha_s \Delta T)$ and $\sigma^{1} = \frac{\sigma}{(1+\gamma_2 \Delta T)}$

$$\frac{h^{1}}{h} = \frac{1 + \gamma_L \Delta T}{(1 + \alpha_s \Delta T)^2} = \frac{1 + \gamma_2 \Delta T}{1 + 2\alpha_s \Delta T}$$

$= 1 + (\gamma_L - 2\alpha_s)\Delta T$

$\Delta h = h^{1} - h = h(\gamma_2 - 2\alpha_s)\Delta T \rightarrow$

From equation (1), $m = hL^2\sigma$

$\rho L^3 = hL^2\sigma$

$$h = \left(\frac{\rho}{\sigma}\right) L = \frac{3.2}{13.6} \times 40$$

$h = 9.41$ cm

$\therefore \Delta h = 9.41[1.8 \times 10^{-4} - 3 \times 10^{-6} \times 2](73)$

$\Delta h = 0.12$ cm

19. $(\Delta v)_g = v\gamma_g \Delta T$ and $(\Delta v)_m = v_m \gamma_m \Delta T$

From given data, $(\Delta v)_g = (\Delta v)_m$

$$v\gamma_g DT = v_m \gamma_m DT \Rightarrow v_m = \frac{v\gamma_g}{\gamma_m}$$

20. If the temperature is considered at a temperature $\theta_2 = \theta_0 + \Delta\theta$

$$L_1^1 = L_0(1 + \alpha_{cu}\Delta\theta)$$
$$L_2^1 = L_0(1 + \alpha_1 \Delta\theta)$$
$$R_1^1 \sim R_2^1 = 2\ d$$
$$\therefore 2\ d(\phi) = L_0(\alpha_{cu} - \alpha_1)(\Delta\theta)$$
$$\phi = \frac{L_0(\alpha_{cu} - \alpha_1)(\Delta\theta)}{2\ d}$$

21. Given, $\gamma = 54 \times 10^{-6}/\ ^{\circ}C$

$$\alpha = \frac{\gamma}{3} = \frac{54 \times 10^{-6}}{3} = 18 \times 10^{-6}/\ ^{0}C$$
$$\beta = 2\alpha = \frac{2}{3}\gamma$$
$$\beta = \frac{2}{3} \times 54 \times 10^{-6} = 36 \times 10^{-6}/\ ^{0}C$$

22. $1 - s, 2 - q, 3 - p, 4 - q, r$

23. We know that, $\gamma = \frac{\Delta\rho}{\rho \Delta T} = \frac{\rho_1 - \rho_2}{\rho \Delta T}$

$$\gamma = \frac{10 - 9.7}{10(100 - 0)} = 3 \times 10^{-4}$$

$\therefore$ Coefficient of linear expansion, $\alpha = \frac{\gamma}{3} = \frac{3 \times 10^{-4}}{3} \Rightarrow \alpha = 1 \times 10^{-4}/\ ^{\circ}C$

24. A bimetallic strip on being heated bends in the form of an arc with more expandable metal (A) outside.

25. Let L_0 be the initial length of each strip before heating.
Length after heating will be

$$L_B = L_0(1 + \alpha_B \Delta T) = (R + d)\theta$$
$$L_C = L_0(1 + \alpha_C \Delta T) = R\theta$$
$$\therefore \frac{R + d}{R} = \frac{1 + \alpha_B \Delta T}{1 + \alpha_C \Delta T}$$
$$1 + \frac{d}{R} = 1 + (\alpha_B - \alpha_C)\Delta T$$
$$R = \frac{d}{(\alpha_B - \alpha_C)\Delta T}$$
$$\therefore R \propto \frac{1}{\Delta T} \text{ and } R \propto \frac{1}{(\alpha_B - \alpha_C)}$$

a. Buoyant force = mg = constan t

$V_{Sub} = \frac{M\gamma}{\rho_1 \gamma}$

∴ Volume of displaced fluid = constant

∴ density of fluid must be constant.

2. $(V_{Solid} - V_{Sub})$ = Constan t

$V_{Solid} - \frac{M_{Solid}}{\rho_{liquid}}$ = Cons tan t

$V \times 3\alpha \times \Delta T = \frac{M\gamma \Delta T}{d}$

$\gamma = 3\alpha\left(\frac{d}{\rho}\right)$

3. $Ah_{in}\, d_{liq} = A(h_{in} + h_{out})\rho_{solid} = M$(mass of Solid)

$h_{out} = \frac{-M}{A\rho_{solid}}$

$h_{in} = \frac{M}{A\rho_{Solid}} - \frac{M}{Ad_{liquid}}$ = constan t

$$\frac{M(1 + 3\alpha\Delta T)}{A(1 + 2\alpha\Delta T)\rho} - \frac{M(1 + \gamma\Delta T)}{A(1 + 2\alpha\Delta T)d} = \frac{M}{A\rho} - \frac{M}{Ad}$$

$\therefore \gamma = 2\alpha + \alpha\frac{d}{\rho}$

4. $A_{in}\, d_{liq}$ = Buoyant force = Cons tan t = $m\gamma$

$A_0 h(1 + 2\alpha\Delta T)h_{in}\left(\frac{d}{1+\gamma\Delta T}\right)$ = Constan t

$h_{in} = \frac{M}{A_o d}[1 + (\gamma - 2\alpha)\Delta T]$

$\therefore \gamma = 2\alpha$

27. When the sphere is floating, at temperature $< 50°$

Weight of body = Thrust

$mg = V_{in}\sigma\gamma \rightarrow (1)$

When body just sinks, at temperature $50°$

$mg = V\sigma^1\gamma$

$\therefore \sigma^1 = \frac{m}{v} = \frac{150}{4/3\pi(2)^3} = 4.48$ g/cm^3

Now, $\sigma^1 = \frac{\alpha}{1+\gamma\Delta T}$

$1 + \gamma\Delta T = \frac{\sigma}{\sigma^1} = \frac{6.5}{4.48}$

$1 + \gamma(50 - 0) = 1.45$

$\gamma(50) = 1 - 1.45 = 0.45$

$\gamma = \frac{0.45}{50}$

$\gamma = 0.00901$

$\gamma = 9.01 \times 10^{-3} \approx \gamma = 9 \times 10^{-3}/{}^{0}\mathrm{C}$

28. Let V_C be the volume of the container and V_1 be the volume of the liquid. According to condition

$\frac{V_1}{V_C}$ = Constant (i.e independent of temperature)

$\frac{V_1^1}{V_C^1} = \frac{V_I}{V_C} \Rightarrow \frac{V_1^1}{V_I} = \frac{V_C^1}{V_C}$

(Here V_I^1 and V_C^1 are volume on heating)

$\frac{V_1^1 - V_1}{V_1} = \frac{V_c^1 V_C}{V_c}$

$\frac{\Delta V_1}{V_I} = \frac{\Delta V_C}{V_C} \Rightarrow \gamma_0 \Delta T = (3\alpha)\Delta T$

$\gamma_0 = 3\alpha$

For container, coefficient of volume expansion will be 3α

29. If on heating H does not change, then the increase in volume of the liquid is accomodated by the increase in base area of the vessel.

Let the area be A.

Initial volume of liquid $= V_1 = A \times H$

Final volume of liquid $= V_1^1 = A^1 \times H$

$\frac{V_1^1}{V_I} = \frac{A^1}{A} \Rightarrow \frac{\Delta V_1^1}{\Delta V} = \frac{\Delta A^1}{A}$

$\gamma_0 \Delta T = \beta \Delta T$

$\gamma_0 = \beta$

$\gamma_0 = 2\alpha (\because \beta = 2\alpha)$

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30. This is exactly same as question no $\gamma_0 = 3\alpha$

WORKSHEET – 3

Level – I

1. A	6. D	10.C	14.C
2. A	7. B	11.B	15. B
3. A	8. A	12.C	
4. D	9. C	13. D	
5. B			

Level – II

1) A	2) C	3) A	4) A	5) B	6) C 7) D
8) C	9) A	10) B	11) B	12) A	13) A 14) D
15) A	16) B	17) A	18) C	19) C	20) D 21) B
22) A	23) A	24) C	25) B	26) B	27) C 28) A
29) a-r,b-s,c-p,d-q		30) 17	31) B	32) C	33) A
34) A,B,C,D					

1. $\gamma_R = \frac{V_2 - V_1}{V_1 \Delta t} / ^\circ C$

$\gamma_R = \frac{80.4 - 80}{80 \times 100} = \frac{0.4}{80 \times 100}$

$\gamma_R = 5 \times 10^{-5} / ^\circ C$

2. Given $\gamma_a = \frac{7}{8}\gamma_R$

We Know that, $\gamma_R = \gamma_a + \gamma_g$

$\gamma_R = \frac{7}{8}\gamma_R + \gamma_g$

$\gamma_R - \frac{7}{8}\gamma_R = \gamma_g$

$\gamma_g = \frac{\gamma_R}{8}$

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3. $\gamma_R = \gamma_a + \gamma_g$

$\gamma_R = \gamma_a + 3\alpha$

$0.000182 = \gamma_a + 0.000027$

$\gamma_a = 0.000182 - 0.000027$

$\gamma_a = 0.000155 / ^\circ C$

4. We know that, $\rho_0 = \rho_t(1 + \gamma_R \Delta t)$

$\rho_t = \frac{\rho_0}{1 + \gamma_R \Delta t}$

given, $\rho_0 = 13.6$ g/cc

$\rho_t = \rho_{200^\circ C} = ?$

$\gamma_R = 0.18 \times 10^{-3} / ^\circ C$

$\Delta t = 200^\circ C$

$\rho_{200} = \frac{13.6}{1 + 0.18 \times 10^{-3} \times 200}$

$\rho_{200} = 13.3 g/cc$

5. Let Volume of the vessel = V

Given, volume at $0^\circ C (V_1) = \frac{1}{6}(V)$

Volume at $200^\circ C (V_2) = \frac{1}{5}(V)$ and $\Delta t = 200^\circ C$

$\gamma_a = \frac{\frac{1}{5}(V) - \frac{1}{6}(V)}{\frac{1}{6}(V)(200)} = \frac{(6V - 5V)}{5 \times (V) \times 200}$

$\gamma_a = \frac{V}{5V \times 200} = \frac{1}{1000}$

$\gamma_a = 1 \times 10^{-3} / ^\circ C$

6. Given $(\gamma_R)_{\text{Gold}} = G + 3A$

$(\gamma_R)_{\text{Silver}} = S + 3\alpha$

$\therefore G + 3A = S + 3\alpha$

$(\alpha)_{\text{Silver}} = \frac{G - S + 3A}{3}$

7. Given, $\gamma_R = \gamma_a + 1\%$ of γ_a

$$\gamma_R = \gamma_a + \frac{\gamma_a}{100}$$
$$\gamma_R = \frac{101\gamma_a}{100}$$
$$\gamma_a = \frac{100\gamma_R}{101}$$

We know that $\gamma_R = \gamma_a + 3\alpha$

$$\gamma_R = \frac{100\gamma_R}{101} + 3\alpha$$
$$\gamma_R - \frac{100\gamma_R}{101} = 3\alpha$$
$$\frac{\gamma_R}{101} = 3\alpha \Rightarrow \alpha = \frac{\gamma_R}{303}$$

8. Volume of the liquid at $t_2^0 c$ is

$$V_2 = V_1[1 + \gamma_R(t_2 - t_1)]$$
$$V_2 = 50[1 + 7 \times 10^{-4} \times 90]$$
$$V_2 = 53.15cc$$

9. Coefficient of apparent expansion

$$\gamma_a = \frac{V_2 - V_1}{V_1(t_2 - t_1)} = \frac{2}{100(110 - 10)} = 2 \times 10^{-4}/\,^\circ\text{C}$$

$\gamma_g = 3\alpha = 3 \times 11 \times 10^{-6} = 33 \times 10^{-6} = 0.33 \times 10^{-4}/\,^\circ\text{C}$

Coefficient of real expansion of the liquid,

$$\gamma_R = \gamma_a + 3\alpha$$
$$\gamma_R = 2 \times 10^{-4} + 0.33 \times 10^{-4}$$
$$\gamma_R = 2.33 \times 10^{-4}/\,^\circ\text{C}$$

10. When the temperature is increased, volume of the cube will increase while density of liquid will decrease. the depth upto which the cube is submerged in the liquid remains the same, hence the uthrust will not change.

$$F = F^1$$
$$V_l \rho_L g = V_l^1 \rho_L^1 g$$
$$(Ah_1)\rho_L g = A(1 + 2\alpha_s \Delta T)h_1 \left(\frac{\rho_L}{1 + \gamma_l \Delta T}\right) g$$

Solving this equation, we get $\gamma_l = 2\alpha_s$

11. $d_1 = 1\text{gm/cc}, d_2 = 0.98\text{gm/cc}$

The block sinks when the density of the block and water are just equal.

$$d_2 = \frac{d_1}{1 + \gamma \Delta t}$$
$$0.984 = \frac{1}{1 + 2.1 \times 10^{-4}(t - 4)}$$

by solving, we get

$$t = 77.43^\circ\text{C}$$

It must be heated to $t^1 = 77.43 + 4 = 81.43^\circ\text{C}$

12. Loss of wt at $0^\circ\text{C} = V_0 \rho_0 g = \omega_0$

Where, V_0 = Volume of body at 0°C

V_1 = Volume of body at $t_1^0 c$

$$\frac{\omega_1}{\omega_0} = \frac{V_1 \rho_1}{V_0 \rho_0} = \frac{d_0}{d_1} \times \frac{\rho_1}{\rho_0}$$

$= 250 \times 1.8 \times 10^{-5} \times 80 \rightarrow (1)$

Expansion of glass container $= V_g \gamma_g \Delta T$

$= 250 \times 3 \times 9 \times 10^{-6} \times 80 \rightarrow (2)$
The volume of mercury overflows will be obtained by subtracting equqation (2) from (1) $\Delta V = 250 \times 80 \times (1.8 - 0.27) \times 10^{-4}$
$\Delta V = 3.06cc$

16. $\Delta l_1 = \frac{\gamma_{Hg} V_{Hg} \Delta T}{A}$ and

$$\Delta l_2 = \frac{\gamma_{Hg} V_{Hg} \Delta T - \gamma_g V_g \Delta T}{A}$$

$$\frac{\Delta l_1 - \Delta l_2}{\Delta l_1} = \frac{\gamma_{Hg} V_{Hg} - (\gamma_{Hg} V_{Hg} - \gamma_g V_g)}{\gamma_{Hg} V_{Hg}}$$

$$\frac{\Delta l_1 - \Delta l_2}{\Delta l_1} = \frac{\gamma_g V_g}{\gamma_{Hg} V_{Hg}} = \frac{3\alpha_g}{\gamma_{1+g}}$$

$(\quad \text{as } V_{1+g} = V_g, \gamma_g = 3\alpha_g)$

$$\frac{\Delta l_1 - \Delta l_2}{\Delta l_1} = \frac{3(8.3 \times 10^{-6})}{182 \times 10^{-6}}$$

$$\frac{\Delta l_1 - \Delta l_2}{\Delta l_1} = 0.14$$

17. Let $\theta = T_2 - T_1$, and $\gamma =$ coefficient of volume expansion of liquid
Let density of liquid at temperatures T_1 and T_2 be ρ_1 and ρ_2 respectively
$\rho_1 = \rho_2(1 + \gamma\theta) \rightarrow (1)$
Let V_1 and V_2 be the volumes of the sinker at temperature T_1 and T_2 respectively.
$V_2 = V_1(1 + \gamma_s\theta) \rightarrow$
The loss in weight at $T_1 = V_1\rho_1 g \Rightarrow \omega_0 - \omega_1 = V_1\rho_1 g \rightarrow (3)$
The loss in weight at $T_2\rho_2 g \rightarrow \omega_0 - \omega_2 = Vl2\rho g \rightarrow (4)$
Dividing equation (3) by equation (4), we get

$$\frac{\omega_0 - \omega_1}{\omega_0 - \omega_2} = \frac{V_1\rho_1}{V_2\rho_2}$$

Using equation (1) and (2), we get

$$\frac{\omega_0 - \omega_1}{\omega_0 - \omega_2} = \frac{1 + \gamma\theta}{1 + \gamma_s\theta}$$

$$\rightarrow 1 + g\theta = \frac{\omega_0 - \omega_1}{\omega_0 - \omega_2} + g_s\left(\frac{\omega_0 - \omega_1}{\omega_0 - \omega_2}\right)\theta$$

$$\gamma = \left(\frac{\omega_2 - \omega_1}{\omega_0 - \omega_2}\right)\left(\frac{1}{T_2 - T_1}\right) + \left(\frac{\omega_0 - \omega_1}{\omega_0 - \omega_2}\right)\gamma_s$$

18. Apparent expansion of mercury $= \frac{\pi}{4}(6 \times 10^{-4})^2 \times 10 \times 10^{-2}$
$= 9\pi \times 10^{-9}$ m^3
Volume of mercury $= \frac{43 \times 10^{-3}}{13.6 \times 1000} = 3.16 \times 10^{-6}$ m^3
As $\Delta V = Vg\Delta T$
$9\pi \times 10^{-9} = 3.16 \times 10^{-6} \times g_a \times 50$
$\gamma_a = 18 \times 10^{-5}$
$\therefore \gamma_R = \gamma_a + \gamma_g = \gamma_a + 3\alpha_g$
$\gamma_R = 18 \times 10^{-5} + 3 \times 9 \times 10^{-6}$
$\gamma_R = 20.7 \times 10^{-5} K^{-1}$

19. From given data,

$$f_1 = V_0(1 + \gamma t_1) \text{ and}$$

$$f_2 = V_0(1 + \gamma t_2)$$
$$\therefore \frac{f_1}{f_2} = \frac{1 + \gamma t_1}{1 + \gamma t_2}$$
$$\gamma = \frac{f_1 - f_2}{f_2 t_1 - f_1 t_2}$$

20. We have $\frac{\Delta V}{V} = \gamma \Delta T = (1.82 \times 10^{-4})(80)$

$$\frac{\Delta V}{V} = 1.46 \times 10^{-2}$$

Now the bulk modulus, P = B $\left(\frac{\Delta V}{V}\right)$ Where P is the pressure.

$$P = (2500)(1.46 \times 10^{-2}) = 36Mpa$$

21. Expansion in mercury = $V_0 \gamma_m T$
Expansion in glass bulb == $V_0(3\alpha)T$
Apparent expansion in mercury = $V_0 \gamma_m T - V_0(3\alpha)T$

$$\therefore A_t l = V_0 T(\gamma_m - 3\alpha)$$
but, $A_t = A_0(1 + 2\alpha T)$
$$\therefore A_0\left(1 + 2\alpha_g T\right) l = V_0 T(\gamma_m - 3\alpha)$$
$$l = \frac{V_0 T(\gamma_m - 3\alpha)}{A_0\left(1 + 2\alpha_g T\right)}$$

22. Volume of glass flask = Volume of mercury + Volume of air.
$V = V_m + V_a$
As V_a remains constant, So $\Delta V = \Delta V_m$
$\gamma_g V \Delta T = \gamma_m V_m \Delta T$

$$V_m = \frac{\gamma_g V}{\gamma_m} = \frac{3\alpha_g V}{\gamma_m}$$
$$V_m = \frac{3 \times 9 \times 10^{-6}}{1.8 \times 10^{-4}} \times 1000$$
$$V_m = 150 \text{ cm}^3$$

23. Let ρ_{F_e} and ρ_{Hg} denote the densities of iron and mercury at 0°C, and m = mass of the block
∴ Volume of the block = $\frac{m}{\rho_{F_e}}$
Now, Weight of mercury displaced = Weight of the block

$$\left(\frac{K_1 m}{\rho_{F_e}}\right)\rho_{Hg} g = mg \Rightarrow K_1 = \frac{\rho_{F_e}}{\rho_{Hg}}$$

Also, $\rho_t = \frac{\rho_0}{1+\gamma t}$

$$\therefore K_2 = \frac{\rho_{F_e}}{1 + \gamma_{F_e} \times 60} \times \frac{1 + \gamma_{\text{Hg}} \times 60}{\rho_{\text{Hg}}} = (K_1)\frac{1 + 60\gamma_{\text{Hg}}}{1 + 60\gamma_{F_e}}$$

$$\frac{K_1}{K_2} = \frac{1 + 60\gamma_{F_e}}{1 + 60\gamma_{Hg}}$$

24. At 50°C, $\rho_s = \rho_l$

$$\frac{V_s}{V_l} = \left(\frac{\rho_s}{\rho_l}\right) at\ ^\circ\text{C}$$

let, $\rho_{50} = \frac{\rho_0}{1+50g}$
$\rho_0 = \rho_{50} = (1 + 50 \text{ g})$

$$\therefore \frac{\rho_s}{\rho_l} = \frac{1 + 50\gamma_s}{1 + 50\gamma_l} = \frac{1 + 50 \times 0.3 \times 10^{-5}}{1 + 50 \times 8 \times 10^{-5}} = 0.9999$$

$\therefore \frac{V_s}{V_l} = 0.9999$

$\left(\frac{V_s}{V_l} \times 100\right)_{at 0^\circ c} = 99.99\%$

25. Let m be the mass of due solid and V its volume by the law of flotation
Weight of floating object=buoyant force.

$$mg = \left(\frac{3}{4}V\right)\rho_{20}g \text{ and } mg\left(\frac{80}{100}V\right)\rho_{100}g$$

$$\therefore \frac{3}{4}\rho_{20} = \frac{4}{5}\rho_{10} \Rightarrow \frac{3}{4}\frac{\rho_0}{(1+g\times 20)} = \frac{4}{5}\frac{\rho_0}{(1+g\times 100)}$$

After solving we get

$$\gamma = 8.47 \times 10^{-4}/^\circ\text{C}$$

26. Water overflow = (final volume of water) - (final volume of bottle)
Water overflow $= 250(1+\gamma_l\theta) - 250$
$= 250 \times 60 \times 10^{-5} \times 10$
$= 1.5 \text{ cm}^3$

27. Water overflow = (final volume of water) - (final volume of glass)

$$= 250(1+\gamma_l\theta) - 250(1+\gamma_g\theta)$$
$$= 250(\gamma_l - \gamma_g)\theta$$

Where $\gamma_g = \frac{3}{2}\beta = 1.8 \times 10^{-5}/^\circ\text{C}$

$$= 250(58.2 \times 10^{-5})(60-50)$$

Water overflow $= 1.455 \text{ cm}^3$

28. $\Delta V_L = \Delta V_L$

$$\gamma_L V_L = \gamma_V V_V \text{ or } \frac{\gamma_L}{\gamma_V} = \frac{V_V}{V_L}$$
$$V_V > V_L \Rightarrow \gamma_L > \gamma_V$$

30. We know that, $\gamma_R = \gamma_a + 3\alpha$

$$\gamma_a + 3\alpha_g = \gamma_a + 3\alpha_m$$
$$10.3 \times 10^{-4} + 27 \times 10^{-6} = 10.06 \times 10^{-4} + 3\alpha_m$$
$$10.57 \times 10^{-4} = 10.06 \times 10^{-4} + 3\alpha_m$$
$$3\alpha_m = 0.51 \times 10^{-4}$$
$$\alpha_m = 0.17 \times 10^{-4}$$
$$\alpha_m = 17 \times 10^{-6}/^\circ\text{C}$$

31. $V_{\text{beaker}} = V_0[1 + 3\alpha_g\Delta t]$

$$V_b = 1[1 + 3 \times 8.3 \times 10^{-6} \times 50]$$
$$V_b = 1.001L$$

32. $V_{\text{Hg}} = V_0[1 + \gamma_{\text{Hg}}\Delta\theta]$
$= 1[1 + 1.82 \times 10^{-4} \times 50]$
$V_{\text{Hg}} = 1.009 \text{ L}$
The overflow is $= 1.009 - 1.001$
$= 0.0008 \text{ L}$
$= 8mL$

33. OPTION -1 (Conceptual)

34. OPTION -1,2,3,4 (Conceptual)

WORKSHEET – 4

Level – I

1. D	5. D	9. C	13. C
2. C	6. D	10. C	14. B
3. A	7. D	11. C	15. B
4. C	8. C	12. D	16. A

12. $V_1 = 1$ litre $t_1 = 27°C$
$V_2 = x = ?$ $t_2 = 177°C$

$$\gamma_p = \frac{1}{273}$$

$$\gamma_p = \frac{V_2 - V_1}{V_1 t_2 - V_2 t_1}$$

$$\frac{1}{273} = \frac{x - 1}{(1)(177) - (x)(27)}$$

x = 1.5 litre

Level – II

1) B	2) D	3) D	4) A	5) C	6) C	7) A
8) C	9) B	10) A	11) D	12) A	13) D	14) D
15) A	16) B	17) B	18) C	19) D	20) B	21) C
22) C	23) B	24) B	25) A	26) B,C	27) A,D	28) A
29) a-r, b-s, c-q, d-p		30) B	31) C	32) x= 4	33) C	34) x= 3
35) D	36) D	37) A	38) ratio = 1	39) D	40) A	

1. We know that,
Volume coefficient $(\gamma_p) = \frac{1}{273} / °C$
Pressure coefficient $(\gamma_v) = \frac{1}{273} / °C$
$\therefore \gamma_p - \gamma_v = \frac{1}{273} - \frac{1}{273}$
$\gamma_p - \gamma_v = 0$

2. At constant pressure, volume coefficient of a gas

$$(\gamma_p) = \frac{V_2 - V_1}{V_1 t_2 - V_2 t_1}$$

$$(\gamma_p) = \frac{1200 - 1000}{1000 \times 87 - 1200 \times 27}$$

$$(\gamma_p) = \frac{200}{54600}$$

$$(\gamma_p) = \frac{1}{273} / °C$$

3. Coefficient of cubical expansion
$\gamma = 3\alpha$

Coefficient of Areal expansion

$\beta = 2\alpha$

given $\gamma = x\beta$

$(3\alpha) = x(2\alpha)$

$x = \frac{3}{2}$

x = 1.5

4. $V_1 = 210cc,\ t_1 = 27°C$

$V_2 = ?\ \ T_1 = 27 + 273 = 300\ K$

By charles law of volume

$\frac{V_1}{T_1} = \frac{V_2}{T_2} \Rightarrow \frac{210}{300} = \frac{V_2}{400}$

$V_2 = \frac{210}{300} \times 400 = 280cc$

5. $V_1 = litre = 1000cc,\ P_1 = 76\ cm\ of\ Hg$

$V_2 = ?\ P_2 = 114\ cm\ of\ Hg$

By Boyle's law ; $P_1\ V_1 = P_2\ V_2$

$V_2 = \frac{P_1 V_1}{P_2} = \frac{76 \times 1000}{114} = 666.7cc$

$V_2 = 667cC$

6. $P_0 = 50\ cm\ of\ Hg, t = 80°C$

$P_t = 64\ cm\ of\ Hg, \gamma_v = ?$

$\gamma_v = \frac{P_t - P_0}{P_0 t}$

$\gamma_v = \frac{64 - 50}{50 + 80} = \frac{14}{4000}$

$\gamma_v = 3.5 \times 10^{-3}$

$\gamma_v = 0.0035/°C$

7. $t_1 = 27°C, P_1 = 75\ cm\ of\ Hg, P_2 = 150\ cm\ of\ Hg$

$\gamma_p = \frac{1}{273},\ t_2 = x = ?$

$\therefore\ \gamma_p = \frac{P_2 - P_1}{P_1 t_2 - P_2 t_1}$

$\frac{1}{273} = \frac{150 - 75}{75 \times x - 150 \times 27} = \frac{75}{75(x - 54)}$

$x - 54 = 273$

$x = 327°C$

8. $t_1 = 0°C, t_2 = 30°C$

$V_1 = 546cc,\ V_2 = x = ?$

$\gamma_p = \frac{1}{273}/°C$

$\gamma_p = \frac{V_2 - V_1}{V_1 t_2 - V_2 t_1}$

$\frac{1}{273} = \frac{x - 546}{546 \times 30 - x \times 0}$

$\frac{1}{273} = \frac{x - 546}{546 + 30}$

$x = 606cc$

Let, $t_1 = 30°C, t_2 = 50°C$

$V_1 = 606cc, V_2 = x^1$

$$\frac{1}{273} = \frac{x^1 - 606}{606 \times 50 - x^1 \times 30}$$

$x^1 = 646cc$

$\therefore$ change in volume $(\Delta V) = V_2^1 - V_2^2$

$$\Delta V = 646 - 606$$

$$\Delta V = 40cC$$

9. $V_1 = 1$ litre, $V_2 = x = ?$

$t_1 = 27°C$, $V_2 = 177°C$

$$\gamma_p = \frac{1}{273}$$

$$\gamma_p = \frac{V_2 - V_1}{V_1 t_2 - V_2 t_1}$$

$$\frac{1}{273} = \frac{x - 1}{1(177) - x(273)}$$

$x = \frac{45}{30} = 1.5$ litre

10. PV = Constant

Case 1:

$P_1 V_1 = P_2 V_2$

$$P_2 = \frac{150P_1}{100} = \frac{3}{2}P_1$$

$$P_1V_1 = \frac{3}{2}P_1V_2$$

$$V_2 = \frac{2}{3} V_1$$

Case 2:

$P_2 V_2 = P_3 V_3$

$$P_3 = \frac{50}{100}P_2 = \frac{1}{2}P_2$$

$$P_2\left(\frac{2}{3}V_1\right) = \frac{1}{2}P_2V_3$$

$$V_3 = \frac{4}{3}V_1$$

11. $P_0 = 54.6$ cm of Hg

$t = 50$

$p_t = ?$

$$\gamma_v = \frac{p_t - p_0}{p_0 t}$$

$$\frac{1}{273} = \frac{p_t - 54.6}{(54.6)50}$$

$p_t = 64.6$ cm of Hg

12. $t_1 = 0°C, t_2 = 100°C, p_0 = 83$ cm, $p_{100} = 113$ cm of Hg

Standard gas thermometer volume is constant

$$\gamma_v = \frac{1}{273}$$
$$\gamma_v = \frac{p_t - p_0}{p_0 t}$$
$$\frac{1}{273} = \frac{143 - 83}{83t}$$
$$t = \frac{60 \times 273}{83} \approx 200°C$$

13. $T_1 = 251\ K, P_1 = 750$ mm of Hg, $P_2 = 1150$ mm of Hg

$$T_2 = ?\ \gamma_v = \frac{p_2 - p_1}{p_1 t_2 - p_2 t_1}$$
$$\frac{1}{273} = \frac{1150 - 750}{750(t_2) - 1150(-22)}$$
$$t_2 = 110°C$$

14. Co - efficient of volume expansion of gas (γ_p) for any type of gas on mixture is ' γ_p '

15. Let initial conditions $= V, T$
and final conditions $= V^1,\ T^1$
By charle's law $V \alpha T$ (is constant)

$$\frac{V}{T} = \frac{V^1}{T^1} \Rightarrow \frac{V}{T} = \frac{V^1}{1.2\ T} \Rightarrow V^1 = 1.2\ V$$

But as per question, volume is reduced by 10% means $V^1 = 0.9\ V$
So percentage of volume leaked out

$$= \frac{(1.2 - 0.9)V}{1.2\ V} \times 100 = 25\%$$

16. According to given data

$$t_1 = 20°C,\ t_2 = 100°C$$
$$V_1 = 100cc,\ V_2 = 125cc$$
$$\gamma_p = \frac{V_2 - V_1}{V_1 t_2 - V_2 t_1}$$
$$\gamma_p = \frac{125 - 100}{100 \times 100 - 125 \times 20} = \frac{25}{7500}$$
$$\gamma_p = \frac{1}{300} / °C$$

17. $t_1 = 27°C, t_2 = ?$
$t_1 = 300\ K, p_1 = 30$ atm $P_2 = 1$ atm

$$V_1 = V, V_2 = 10\ V$$
$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$
$$\frac{30\ V}{300} = \frac{1(10\ V)}{T_2}$$
$$T_2 = 100\ K$$
$$T_2 = 100 - 273$$
$$t_2 = -173°C$$

18. According to Boyle's law, $\frac{p}{d}$ = constan t

$$\therefore V_1 = \frac{m_1}{d_1} = \frac{Km_1}{p_1} \text{ and } V_2 = \frac{Km_2}{P_2}$$

Total volume $= K\left(\frac{m_1}{p} + \frac{m_2}{p_2}\right)$
Let mixture has common pressure p and common density d.

$$p = kd = \frac{(m_1 + m_2)}{\left(\frac{m_1}{p_1} + \frac{m_2}{p_2}\right)}$$

$$p = \frac{p_1 p_2 (m_1 + m_2)}{p_2 m_1 + m_2 p_1}$$

19. Initial pressure of the air $(p_1) = 1$ atm
Pressure pressure of the air $(p_2) = 4$ atm
Initial density of the air $(d_1) = 0.9\text{kgm}^{-3}$
Final density of the air $(d_2) = ?$
From Boyle's law

$$\frac{p_1}{d_1} = \frac{p_2}{d_2} \Rightarrow d_2 = \frac{p_2}{p_1}\, d_1 = \frac{4}{1} \times 0.9 = 3.6$$

$$\therefore d_2 = 3.6\text{kgm}^{-3}$$

20. Initial pressure of the gas $(p_1) = p$
Final pressure of the gas $(p_2) = \frac{110}{100}p$
Initial temperature $t_1^0\text{C} = 27°\text{C}$
$T_1 = 27 + 273 = 300$ K
Final temperature $t_2^0\text{c} = ?$
At constant volume, according to charles law

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

$$T_2 = \frac{p_2}{p_1}\, T_1 = \frac{110p}{100 \times p} \times 300 = 300 \text{ K}$$

$$t_2 = 300 - 273 = 57°\text{C}$$

21. $t_1 = 100°\text{C}, t_2 = x = ?$
$p_1 = 2$ atm, $p_2 = 3$ atm

$$\gamma_v = \frac{1}{273}$$

$$\gamma_v = \frac{p_2 - p_1}{p_1 t_2 - p_2 t_1}$$

$$\frac{1}{273} = \frac{3 - 2}{2x - 300}$$

$$2x - 300 = 273$$

$$2x = 573$$

$X = 286.5°\text{C}$

22. By keeping the pressure of the gas remains constant. According to Charle's law $V_\alpha T$. In the graph is pressure kept constant and volumes are compared.
From charle's law at constant pressure $V_\alpha T$
From the graph $V_2 > V_1$ so $T_2 > T_1$

23. Let p be the atmospheric pressure. When the open and of the tube is up,
$p_1 = p + 15$
When the tube invested,
$p_2 = p - 15$
Using Boyle's law for constant temperature $p_1 v_1 = p_2 v_2$
$(p + 15)(A \times 30) = (p - 15)(A \times 45)$
p = 75 cm of Hg

24. According to Boyle's law = (p_1v_1) at the top lake = (p_2v_2) at the bottom of lake

$$p_1v_1 = (p_1 + h)v_2$$
$$10 \times \frac{4}{3}\pi\left(\frac{5r}{4}\right)^3 = (10 + h)\frac{4}{3}\pi r^3$$
$$\therefore h = 9.53 \text{ m}$$

25. Using pressure or Gay - Lussac's law

$$\frac{p_1}{p_2} = \frac{T_1}{T_2}$$
$$p_2 = \frac{T_1}{T_2}p_1 = \frac{p(273 + 927)}{(273 + 27)} = 4p$$
$$p_2 = 4p$$

26. From the basic concept options (2) and (3) are correct.

27. $t_1^0C = 20°C, t_2 = 100°C$

$$v_1 = 75cc, v_2 = 100cc$$
$$\gamma_p = \frac{v_2 - v_1}{v_1t_2 - v_2t_1}$$
$$\gamma_p = \frac{100 - 75}{75 \times 100 - 100 \times 20} = \frac{25}{5500}$$
$$\gamma_p = \frac{1}{220}/°C(\text{ or }) \alpha = 0.004545/°C$$
$$\gamma_p = 45 \times 10^{-4}/°C$$

30. $v_1 = 4$ litres $v_2 = ?$
$p_1 = 100Nm^{-2}$ $p_2 = 150Nm^{-2}$
Using Boyle's Law for constant temperature
$p_1v_1 = p_2v_2$
$v_2 = \frac{pv_1}{p_2} = \frac{100 \times 4}{150} = 2.667$ litre
change in volume $(\Delta V) = V_1 - V_2$
$\Delta V = 4 - 2.667$
$\Delta V = 1.33$ litre

31. $T_1 = 27 + 273 = 300$ K
$p_1 = 150Nm^{-2}$, $T_2 = 127 + 273 = 400$ K
$p_2 = ?$
Using Gay Lussac's law of constant volume
$$\frac{p_2}{p_1} = \frac{T_2}{T_1}$$
$$p_2 = \frac{T_1}{T_2} \times p_1 = \frac{400 \times 150}{300}$$
$$p_2 = 200Nm^{-2}$$

32. $p \times \frac{1}{v} \Rightarrow \frac{v_2}{v_1} = \frac{p_1}{p_2}$
$$\frac{v_2}{v_1} = \frac{100}{105}$$
$$v_2 = \frac{100}{105}v_1 = 0.953v_1$$
% change in volume $= \frac{V_1 - V_2}{V_1} \times 100$
$$= \frac{V_1 0.953v_1}{v_1} \times 100$$
$$= 4.76\%$$

$= (4 + 0.76)\%$

33. $t_1 = 27°C\ v_1 = v$

$T_1 = 27 + 273 = 300\ K\ v_2 = 3v$

$T_2 = ?$

$\frac{v_1}{T_1} = \frac{v_2}{T_2}$

$\frac{v}{300} = \frac{3v}{T_2}$

$T_2 = 900\ K$

$t_2 = 900 - 273 = 627°C$

$t_2 = 627°C$

34. $t_1 = 0°C\ p_1 = 30$ cm of Hg

$t_2 = 90°C\ p_2 = 72$ cm of Hg

$\gamma_v = \frac{72 - 30}{30 \times 90} = \frac{42}{2700} = 0.01515$

$\gamma_v = 15.15 \times 10^{-13}/°C\ \therefore x = 3$

35. $PV = nRT \Rightarrow nt = cons\ tan\ t$

$n^1(273 + 40) = n(273 + 20)$

$\frac{n^1}{n} = \frac{293}{313} = 0.94$

Fraction escaped $= 1 - 0.94 = 0.06$

36. $V \alpha T$

$V = nT \rightarrow (1)$

$\frac{90}{100}\ V = n^1 \frac{6\ T}{5}$

$\frac{n^1}{n} \times \frac{6}{5} = \frac{9}{10} \Rightarrow \frac{n^1}{n} = \frac{9}{10} \times \frac{5}{6} = \frac{3}{4}$

Percentage of gas leaked out

$$\frac{n - n^1}{n} \times 100 = \frac{4 - 3}{4} \times 100 = 25\%$$

37. OPYION - 1 (Conceptual)

38. For any gas γ_p and γ_v value is same i.e., $\frac{1}{273}/°C$

39. $\Delta V = V - V^1$

$V^1 = V - \Delta V = V - 0.10 = 0.9\ V$

$\therefore p^1 v^1 = pv, p^1 = \frac{pv}{v_1} = \frac{pv}{0.9v} = 1.111p$

Percentage increase in pressure

$$\frac{1.111p - p}{p} \times 100 = 11.1\%$$

40. As $pv = RT$, $p_1 v = 250R$, $p_2 v = 251R$

$\frac{p_2}{p_1} = \frac{251}{250}$

Perecentage increase in pressure $\frac{p_2 \times p_1}{p_1} \times 100$

$= \left(\frac{p_2}{p_1} - 1\right) \times 100$

$= \left(\frac{251}{250} - 1\right) \times 100$

$= \frac{1}{250} \times 100 = 0.4\%$

8. WAVES AND SOUND

WORKSHEET – 1

Level – I

1. A
2. C
3. A
4. A
5. D
6. D
7. D
8. A
9. B

Level – II

1) B	2) B	3) C	4) D	5) D	6) B	7) A
8) C	9) A	10) C	11) A	12) B	13) C	14) B
15) A	16) B	17) B	18) C	19) A	20) A	21) B
22) A,B,C,D	23) A	24) a-s; b- r; c- q; d-p		25) 20	26) B	
27) C	28) 2,3,4	29) a-p; b-p; c-p; d-r		30) 25	31) a-r; b-q; c-q; d-p	

1. The reciprocal of time period is called frequency.
2. One vibration per second is called hertz.
3. The distance travelled by the wave in one time period is called the Wave length of the wave.
4. In constructive interference phase difference is 0° or $2n\pi$
5. $V = \frac{\lambda}{\mathrm{T}}$
6. $f = \frac{1}{T} = \frac{1}{10^{-15}} = 10^{15}$ Hz
7. $f = \frac{1}{T} = \frac{1}{10^{-2}} = 100$ Hz
8. $V = n\lambda$
   $330 = n(66)$
   $n = 5$ Hz
9. $V = n\lambda$
   $330 = \lambda(30)$
   $\lambda = 11$ m
10. $V = n\lambda$
    $3 \times 10^8 = \lambda(10^9)$
    $\lambda = 0.3$ m $= 30$ cm
11. In constructive interference, the resultant amplitude at the point of observation will be maximum.
12. In constructive interference, the resultant amplitude at the point of observation will be minimum.
13. $f = \frac{256}{2} = 128$ Hz
14. Number of vibrations in a minute $= 100 \times 60 = 6000$
15. $A = a_1 + a_2$
16. $A = a_1 - a_2$
17. $\mathrm{T} = \frac{1}{20} = 0.05$ s
18. $V = n\lambda$
    $440 = \lambda(220)$
    $\lambda = 2m$
19. $V = n\lambda$

$$343.4 = n(1.7 \times 10^{-2})$$
$$\text{n} = 2.02 \times 10^{4} \text{ Hz}$$

20. option - 1,2,3,4 (conceptual)
21. option -4 (conceptual)
22. a-s; b-r ; c-q; d-p. (conceptual)
23. $\text{f} = \frac{1}{\text{T}} = \frac{1}{0.05} = 20$ Hz
24. $V = n\lambda$
$$320 = n(1.6 \times 10^{-2})$$
$$n = 20{,}000Hz$$
25. $n_1\lambda_1 = n_2\lambda_2$
$$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{3}{1}$$
26. $\text{y}_1 = 4\sin(1026\pi\text{t})$ and $\text{y}_2 = \sin(1014\pi\text{t})$
$\omega_1 = 1026\pi; \omega_2 = 1014\pi$
$2\pi n_1 = 1026\pi; 2\pi n_2 = 1014\pi$
$n_1 = 513$ Hz; $n_2 = 507$ Hz
beat frequency $= n_1 - n_2$
$$= 513 - 507$$
$$= 6Hz$$
The amplitude of the resulting wave varies with the frequency 3 Hz.
$$\frac{I_{\max}}{I_{\min}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2}$$
$$\frac{I_{\max}}{I_{\min}} = \frac{(4+1)^2}{(4-1)^2}$$
$$\frac{I_{\max}}{I_{\min}} = \frac{25}{9}$$
27. $T = 2 \times 10^{-3}$ s
$V = n\lambda$
$$v = \frac{\lambda}{\text{T}}$$
$$1250 = \frac{\lambda}{2 \times 10^{-3}}$$
$$\lambda = 2500 \times 10^{-3}$$
$$\lambda = 25 \times 10^{-1} \text{ m}$$
28. $a - r; b - q; c - q; d - p$ (conceptual)

WORKSHEET – 2

Level – I

1. D	4. D	6. C	8. D
2. C	5. D	7. D	9. B
3. D			10. A

Level – II

1) B	2) A	3) D	4) D	5) A	6) C	7) B
8) C	9) C	10) A	11) C	12) D	13) B	14) B
15) C	16) A,B,C	17) A	18) a-p; b-p; c-q; d-q		19) 71	20) A
21) B	22) C	23) A,B,C,D		24) C	25) A	26) B
27) C	28) a-p; b-q; c-r; d-s					

1. $n_P - n_Q = 3$ (or) $n_Q - n_P = 3$
Let when P is loaded with wax,
its frequency decreases by x
$(n_P - x) - n_Q = 5$(or)$n_Q - (n_P - x) = 5$
$(n_P - n_Q) - x = 5$(or)$(n_Q - n_P) + x = 5 - - -$ (l)
$3 - x = 5$(or)$3 + x = 5$
$x = 2$(or)$x = 2$
put $x = 2$ in eq (1)
$n_Q - 432 + 2 = 5$
$n_Q = 435$ Hz

3. $\frac{n'}{n} = \frac{11}{12}$

$$n' = n\left(\frac{V}{V + V_s}\right)$$

$\frac{11}{12} = \frac{V}{V + V_s}$
$11\ V + 11\ V_s = 12\ V$
$V_s = \frac{V}{11}$

4. $n' = n\left(\frac{V}{V+V_s}\right)$
$\frac{n'}{n} = \frac{330}{330 + 110}$
$\frac{n'}{n} = \frac{330}{440} = \frac{3}{4}$

5. $n' = n\left(\frac{V-V_0}{V+V_s}\right)$
$n' = 1100\left(\frac{350 - 200}{350 + 200}\right)$
$n' = 300$ Hz

6. $n_A - n_B = 5$ (or) $n_B - n_A = 5$
Let when A is loaded with wax,
its frequency decreases by x
$(n_A - x) - n_B = 2$ (or) $n_B - (n_A - x) = 2$
$(n_A - n_B) - x = 2$(or)$(n_B - n_A) + x = 2 - - \cdot$
$5 - x = 2$(or)$5 + x = 2$
$x = 3$ (or) $x = 3$
put $x = 3$ in eq(1)
$(n_A - n_B) - x = 2$
$250 - n_B - 3 = 2$
$n_B = 245$ Hz

7. $200 - x = 4$ (or) $x - 200 = 4$
$x = 196Hz$(or)$x = 204Hz$

8. $n_1 = 21x; n_2 = 20x$
$n_1 - n_2 = 5$
$21x - 20x = 5$
$x = 5$
$n_1 = 105$ Hz; $n_2 = 100$ Hz

9. $n' = n\left(\frac{V}{v+V_s}\right)$
$n' = 480\left(\frac{340}{340 - 20}\right)$

$n' = 480\left(\frac{340}{320}\right)$

$\mathrm{n}' = 510$ Hz

$n' - n = 510 - 480 = 30$ Hz

10. $\mathrm{n}' = \mathrm{n}\left(\frac{\mathrm{V}-\mathrm{V}_0}{\mathrm{V}+\mathrm{V}_s}\right)$

$1950 = n\left(\frac{340-10}{340+10}\right)$

$1950 = n\left(\frac{330}{350}\right)$

$\mathrm{n} \approx 2068$ Hz

11. $\mathrm{n} = 100; \mathrm{n}' = 110$ Hz

$$n' = n\left(\frac{V}{V - V_s}\right)$$

$$110 = 100\left(\frac{330}{330 - V_s}\right)$$

$$\frac{330 - V_s}{330} = \frac{10}{11}$$

$11V_s = 330$

$V_s = 30$ m/s

12. $\overrightarrow{V_0} = \sqrt{2}(\hat{\imath} + \hat{\jmath})$

$\mathrm{V}_0 = 2$ m/s

If the observer is moving away from the source

$$n' = n\left(\frac{V - V_0}{V}\right)$$

$$n' = 680\left(\frac{340 - 2}{340}\right)$$

$n' = 676$ Hz

If the observer is moving towards the source

$$n' = n\left(\frac{V + V_0}{V}\right)$$

$$n' = 680\left(\frac{340 + 2}{340}\right)$$

$n' = 684$ Hz

13. $n' = n\left(\frac{V}{V+V_s}\right)$

$970 = 1000\left(\frac{330}{330 + V_s}\right)$

$V_s = 10$ m/s

The wall will act as source

$$n' = n\left(\frac{V}{V - V_s}\right)$$

$$n' = 1000\left(\frac{330}{320}\right)$$

$n' \approx 1032Hz$

14. $\mathrm{n}' = \mathrm{n}\left(\frac{\mathrm{V}-\mathrm{V}_0}{\mathrm{V}}\right)$

$\mathrm{n} = 100; \mathrm{n}' = 95$

$95 = 100\left(\frac{340 - V_0}{340}\right)$

$323 = 340 - \mathrm{V}_0$

$V_0 = 340 - 323$

$V_0 = 17$ m/s

15. $n' = n\left(\frac{V+V_0}{V-V_s}\right)$

$n' = 600\left(\frac{340+15}{340-40}\right)$

$n' = 710$ Hz.

16. OPTION - 1,2,3 (Conceptual)

17. OPTION - 1 (Conceptual)

18. a)

P is waxed, Number of beats decreased

therefore, frequency Q is less.

$280 - n_Q = 6$

$n_Q = 280 - 6$

$n_Q = 274$ Hz

b) Q is filed, Number of beats decreased therefore, frequency Q is less.

$280 - n_Q = 6$

$n_Q = 280 - 6$

$n_Q = 274$ Hz

c) P is filed, number of beats remain same.

therefore, frequency Q is more.

$n_Q - 280 = 6$

$n_Q = 286$ Hz

d) Q is filed, and number of beats is increased

therefore, frequency Q is more.

$n_Q - 280 = 6$

$n_Q = 286$ Hz

19. When source is moving towards stationary observer $n' = n\left(\frac{V}{V-V_s}\right)$

20. When source is moving away from the stationary observer $n' = n\left(\frac{V}{V+V_s}\right)$

21. When observer is moving towards the stationary source

$n' = n\left(\frac{V+V_0}{V}\right)$

22. OPTION - 1,2,3,4 (conceptual)

23. OPTION - 3 (conceptual)

24. When observer and source are moving towards each other. $n' = n\left(\frac{V+V_0}{V-V_s}\right)$

25. When observer and source are moving away from each other.$n' = n\left(\frac{V-V_0}{V+V_s}\right)$

26. When source is moving behind the observer $n' = n\left(\frac{V-V_0}{V-V_s}\right)$

27. a) When source is moving towards stationary observer

$$n' = n\left(\frac{V}{V-V_s}\right); \lambda' = \lambda\left(\frac{V-V_s}{V}\right)$$

b) When source is moving away from the stationary observer

$$n' = n\left(\frac{V}{V+V_s}\right): \lambda' = \lambda\left(\frac{V+V_s}{V}\right)$$

c) When observer and source are moving towards each other.

$$n' = n\left(\frac{V+V_0}{V-V_s}\right); \lambda' = \lambda\left(\frac{V-V_s}{V+V_0}\right)$$

d) When observer and source are moving away from each other.

$$n' = n\left(\frac{V-V_0}{V+V_s}\right); \lambda' = \lambda\left(\frac{V+V_s}{V-V_0}\right)$$

9. RAY OPTICS

WORKSHEET – 1

Level – I

1.B 2.D 3.A 4.A 5.C 6.A

Level – II

1) B 2) C 3) A 4) A 5) A 6) B 7) A

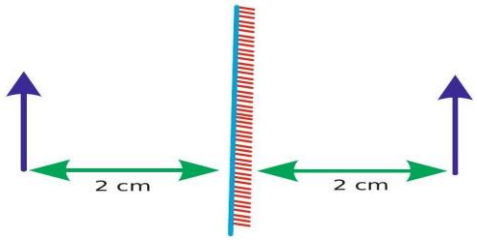
8) C 9) D 10) D 11) D 12) C 13) C 14) A

15) A 16) A,B 17)a- 3; b- 4; c -1;d- 2 18) 10 cm 19) C 20) A

21) A 22) A 23) A

1. Angle B is the angle of incidence (angle between the incident ray and the normal).
2. Angle C is the angle of reflection (angle between the reflected ray and the normal).
3. The angle of reflection is 60 degrees. (Note that the angle of incidence is not 30 degrees; it is 60 degrees since the angle of incidence is measured between the incident ray and the normal.)
4. $\delta = 180^\circ - 2\mathrm{i} = 180^\circ - 2 \times 60^\circ = 60^\circ$
5. The light reflects twice before it finally exits the system. Draw a normal at the point of incidence to the first mirror; measure the angle of incidence (45 degrees); then draw a reflected ray at 45 degrees from the normal. Repeat the process for the second mirror.
6. If the light makes an angle of 10° with the surface, it makes an angle of 80° with the normal to the surface. Thus the angle of incidence is 80°.
7. According to the law of reflection, the angle of reflection equals the angle of incidence. So the angle of reflection (measured to the normal) is 80°
8. According to the law of reflection, the angle of reflection must equal the angle of incidence for each ray. Thus ray A will have an angle of reflection of 80°.
9. U, V, W, not does not show lateral inversion.
10. Only letter Z shows lateral inversion..
11. Conceptual
12. Size of mirror is equal to half of height of person. i.e., 91 cm
13. Distance between object and mirror is equal to distance between mirror and image. Image suffers a displacement of 6 cm in the opposite direction.
14. Size of object infront of mirror =size of image formed inside the mirror. size of image = 5 cm.
15. Light is a form of energy produce by luminous object
16. A, X does not show lateral inversion
17. Conceptual.
18. Size of object infront of mirror =size of image formed inside the mirror. size of image = 10 cm
19. Parallel.
20. Image formed inside the mirror is virtual.
21. Image formed inside the mirror is virtual

22. Virtual images Cannot be caught on screen.
23. $\angle i = \angle r; \angle r = 15°$
24. $\angle i = \angle r; \angle r = 31°$
25. $\angle i = \angle r; \angle r = 47^0$
26. Virtual images are always erect.
27. Real image
28. Real image can be caught on screen.
29. Size of mirror is equal to half of height of person. i.e., 20 cm
30. Size of mirror is equal to half of height of person. i.e., 90 cm



Distance of the image from the man is $2 + 2 = 4$ cm

WORKSHEET – 2

Level – I

1) A 2) A 3) A 4) C

Level – II

1) C	2) A	3) C	4) C	5) C	6) C
7) A	8) A	9) B	10) A	11) B	12) C
13) A	14) C	15) B	16) C	17) A	18) A
19) A	20) D	21) A	22)1-r; 2-p; 3-q; 4-t	23) B	24) A
25) C	26) C	27) C	28) C		

1. $\delta = 180° - 2i$; $\delta = 180° - 2 \times 45° = 180° - 90° = 90°$
2. The number of images formed when two mirrors are inclined to each other is given by $n = (360/\theta - 1)$
   here $\theta = 90°$ [sin cewalls areperpendicular] $= 4 - 1 = 3$
3. The reflected ray rotates through an angle 2θ.
4. Size of the image will not change.
5. $\theta = 50°$
   $n = \frac{360°}{\theta} = \frac{360°}{50°} = 7.2$ is odd
   so no, of images = 7
6. $\theta = 60°$
   $n = \frac{360°}{\theta} = 6$ is even
   so no of images are $6 - 1 = 5$
7. From laws of reflection $|i = |$; from question $i = 0$ so $r = 0$.
8. $\delta = 60°; \delta = 180° - 2i$
   $60° = 180° - 2i$
   $2i = 180° - 60° = 120°$; $i = \frac{120°}{2} = 60°$
   glancing angle $= 90° - 60° = 30°$.
9. Angle of deviation from two mirrors
   $\delta = 360° - 2\theta$
   $200° = 360° - 2\theta$
   $2\theta = 360° - 200°$

$2\theta = 160^\circ$

$\theta = \frac{160^\circ}{2} = 80^\circ$ is angle b/w mirrors

10. i = angle of incidence. angle of deviation $\delta = 2i$

from formula $\delta = 180^\circ - 2i$

$$2i = 180^\circ - 2i;; 4i = 180^\circ; i = \frac{180^\circ}{4} = 45^\circ$$

11. From problem $|i = 0$, formula of deviation

$$\delta = 180^\circ - 2i = 180^\circ - 2 \times 0^0 = 180^\circ$$

12. i - angle of incidence

r - angle of reflection

Given $i + r, r = x$

from (1)

$x + x = 90^\circ$

$2x = 90^\circ$

$x = 45^\circ$

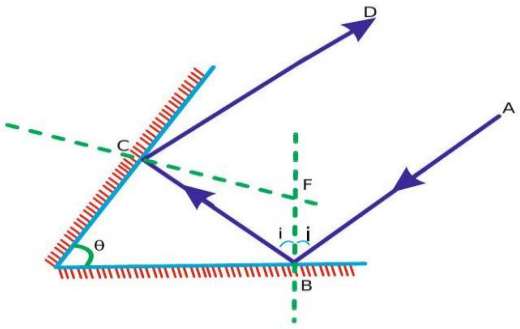
So angle of incidence $= 45^\circ, r = 45^\circ$ also

Angle of deviation $\delta = 180^\circ - 2i = 180^\circ - 2 \times 45^\circ = 180^\circ - 90^\circ = 90^\circ$

13. $i = 30^\circ$

$\delta = 180^\circ - 2i = 180^\circ - 2 \times 30^\circ = 180^\circ - 60^\circ = 120^\circ$

14.



AB = ncident ray and θ = angle at incidence $AB \parallel CD$

$$\therefore \underline{\text{DCB}} = 180^\circ - 2\theta \begin{cases} \text{AB} \parallel \text{CD} \\ \underline{\text{ABC+| DCB} = 180^\circ} \end{cases}$$

$\underline{BCF} = \frac{180^\circ - 2\theta}{2} - 90^\circ - \theta$

Now in $\triangle BFC$

$\therefore \angle\text{BFC} + \angle\text{FBC} + \angle\text{FCB} = 180^\circ$

$\therefore \angle\text{BFC} + \theta + 90^\circ - \theta = 180^\circ$

$\therefore \angle\text{BFC} = 90^\circ$

Now in quadrilateral

$\underline{F} + \underline{C} +$

$|\theta = 360^\circ - 270^\circ = 90^\circ$

15. $\delta = $ i from problem

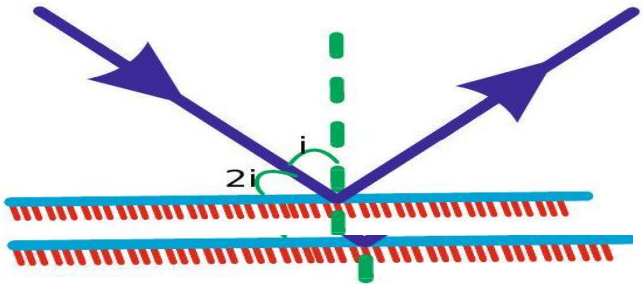
$\delta = 180^\circ - 2i$

$3i = 180^\circ$

$i = 60^\circ$

if $i = 60^\circ$ Then glancing angle of incidence is 30°.

16.



$i + 2i = 90°$
$3i = 90°$
$i = 30°$
$\delta = 180° - 2i$
$\delta = 180° - 2(30°)$
$\delta = 120°$

17. Angle of deviation produced by 2 mirrors is $\delta = 2\pi - 2\theta$ formula.
18. $\delta = 2\pi - 2\theta = 2 \times 180° - 2 \times 30° = 360° - 60° = 300°$
19. $\delta = 300° \theta = ?$
$\delta = 360° - 2\theta$
$300° = 360° - 2\theta$
$2\theta = 360° - 300° = 60°$
$\theta = 30°$
20. $\delta = 360° - 2\theta = 360° - 2 \times 60°$
$\delta = 360° - 120° = 240°$
21. Angle b/w the reflected and incident ray $= 80°$
angle of incidence = angle of reflected ray $= 40°$
angle of reflected ray with mirror $= 90° - 40° = 50°$
so, angle with horizontal $= 90° - 50° = 40°$

a. $n = \frac{360°}{60°} = 6 -$ even so $6 - 1 = 5$
2. $n = \frac{360°}{45°} = 8 -$ even so $8 - 1 = 7$
3. $n = \frac{360°}{90°} = 4 -$ even so $4 - 1 = 3$
4. $n = \frac{360°}{20°} = 18 -$ even sol $-1 = 17$

23. Conceptual.
24. $i = 60°$
$r = 60°$
glancingangleof reflection $= 90° - 60° = 30°$
25. glancing angle $= 30°$
From definition angle of incidence $i = 60°$
angle of deviation
$\delta = 180° - 2i$
$= 180° - 2 \times 60°$
$= 180° - 120° = 60°$
26. If mirror is rotated by 20° reflected ray is rotated by 40°
27. If mirror is rotated through an angle θ the reflected ray turns through an angle twice. we are rotating the mirror through an angle 30°. So the re flected ray will rotate in the above manner by an angle 60° How ever you have also rotated incidence ray by 10°. If it is in the same direction as the turning of mirror, this will reduce the angle of incidence by 10°60 the

actual rotation of the reflected ray will be reduced by $10°, 60$ finally you will get the reflected ray turned through $60° - 10° = 50°$

28. Here, $[n] = 5 \Rightarrow n - 1 \leq 5 \leq n$
$\therefore \frac{360}{\theta} - 1 \leq 5 \leq \frac{360}{\theta}$
or, $\theta \geq \frac{360}{6}$ or $\theta \leq \frac{360}{5}$
$\therefore 60° \leq \theta \leq 72°$

WORKSHEET – 3

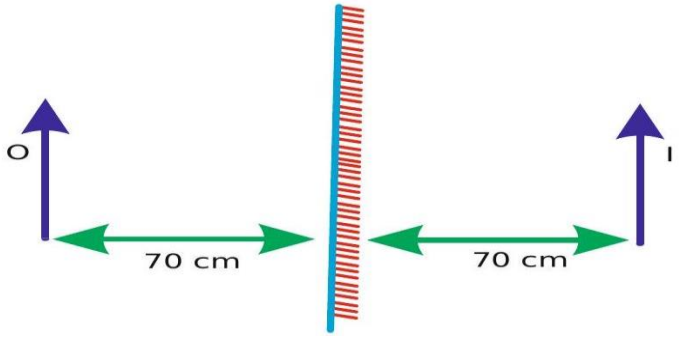
Level – I

1) A 2) B 3) A 4) B

Level – II

1) D	2) D	3) A	4) A	5) A	6) A	7) A
8) B	9) A	10) D	11) C	12) A	13) A	14) B
15) C	16) D	17) A	18) C	19) 1-a;2-b;3-d;4-c		20) X m/s
21) A	22) A	23) A	24) A	25) A	26) D	27) A
28) B	29) C					

1. $V_{IM} = -V_{OM}$
$V_{IM} = -5$ m/s
2. $V_{IO} = -2$ V
$V_{IM} = -2(6)$m/s
$V_{IM} = -12$ m/s
3. $V_{IM} = -V_{OM}$
$V_{IM} = -20$ m/s
4. $V_{IO} = V_{IM} - V_{OM}$
$V_{\mathrm{IO}} = (-1) - (1); V_{\mathrm{IO}} = -2m/s$
5. $V_{IO} = V_{IM} - V_{OM}$
$V_{\mathrm{IO}} = (5) - (-5)$
$V_{\mathrm{IO}} = 10$ m/s
6. $V_{IO} = V_{IM} - V_{OM}$
$-1 = (V) - (-V)$
$V = -\frac{1}{2}$
$|V| = 0.5$ m/s
7. $|V_{IM}| = |V_{OM}|$
$|V_{IM}| = X$
8. After 6 s the mirror covers a distance of 30 cm.
Now, distance between the object and mirror is 70 cm



The distance between object and image = 70 + 70 = 140 cm.

9. Distance between object and mirror after 25sec = 8 + (6 × 25) = 158 cm

By the property of image formed by plane mirror, image distance is equal to object distance. So the distance between object and image after 25 s $= 158 \times 2 = 316$ cm

10. $V_{IO} = V_{IM} - V_{OM}$
$V_{\mathrm{IO}} = -V - V$
$V_{\mathrm{IO}} = -2V = -2(1.2)$
$V_{\mathrm{IO}} = -2.4$ m/s
$|V_{\mathrm{IO}}| = 2.4$ m/s

11. $|V_{IO}| = 2V$
$|V_{IO}| = 2(2)$
$|V_{\mathrm{IO}}| = 4$ m/s

12. $|V_{IO}| = 2V$
$|V_{IO}| = 2(15)$
$|V_{IO}| = 30$ m/s

13. $|V_{IO}| = 2V$
$|V_{IO}| = 2(1)$
$|V_{IO}| = 2$ m/s

14. $|V_{IO}| = 2V$
$|V_{IO}| = 2(2.5)$
$|V_{\mathrm{IO}}| = 5$ m/s

15. $V_{IO} = V_{IM} - V_{OM}$
$V_{IO} = -V - V$
$V_{\mathrm{IO}} = -2V = -2(10)$
$V_{\mathrm{IO}} = -20$ m/s
$|V_{\mathrm{IO}}| = 20$ m/s

16. $|V_{IO}| = 2$ V
$|V_{IO}| = 2(10)$
$|V_{\mathrm{IO}}| = 20$ cm/s

17. $|V_{IM}| = |V_{OM}|$
$|V_{IM}| = 12$ m/s

18. $|V_{IM}| = |V_{OM}|$
$|V_{IM}| = 30$ m/s

19.

a) $V_{IO} = V_{IM} - V_{OM}$
$V_{\mathrm{IO}} = -P - P$
$V_{\mathrm{IO}} = -2P$
$|V_{IO}| = 2P$

b) $V_{IO} = V_{IM} - V_{OM}$
$V_{IO} = -10 - 10$
$V_{IO} = -20$
$|V_{\mathrm{IO}}| = 20$cmls

c) $V_{IO} = V_{IM} - V_{OM}$
$V_{IO} = -X - X$
$V_{\mathrm{IO}} = -2X$
$|V_{IO}| = 2X$cmls

d) $|V_{IO}| = 2X$cmls
$2 = -V - V$
$-2V = 2$
$V = -1$
$|V| = 1$ m/s

20.

$$|V_{IM}| = |V_{OM}|$$
$$|V_{IM}| = X \text{ m/s}$$

21.

$$V_{IO} = V_{IM} - V_{OM}$$
$$V_{IO} = -0.5 - 0.5$$
$$V_{IO} = -1$$
$$|V_{IO}| = 1\text{mls}$$

22.

$$V_{IO} = V_{IM} - V_{OM}$$
$$V_{IO} = -\left(\frac{V}{2}\right) - \left(\frac{V}{2}\right)$$
$$V_{IO} = -V$$
$$|V_{IO}| = V\text{mls}$$

23.

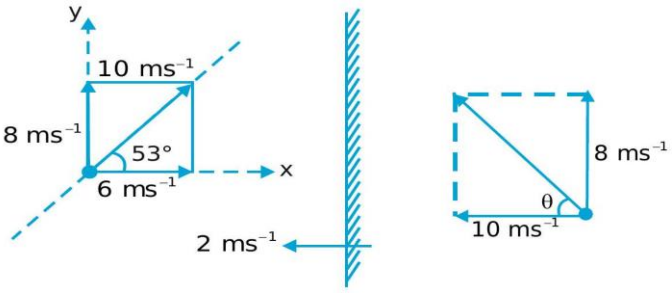
$$V_{IO} = V_{IM} - V_{OM}$$
$$V_{IO} = -\left(\frac{V}{4}\right) - \left(\frac{V}{4}\right)$$
$$V_{IO} = -\left(\frac{V}{2}\right)$$
$$|V_{IO}| = \left(\frac{V}{2}\right)\text{mls}$$

24.

$$V_{IO} = V_{IM} - V_{OM}$$
$$V_{IO} = -\left(\frac{V}{6}\right) - \left(\frac{V}{6}\right)$$
$$V_{IO} = -\left(\frac{V}{3}\right)$$
$$|V_{IO}| = \left(\frac{V}{3}\right)\text{mls}$$

25. When the object moves infront of stationary mirror, the relative speed between object and its image along the plane of the mirror is zero

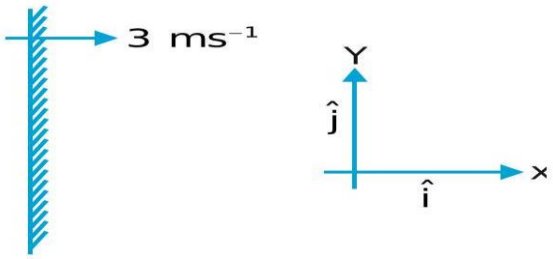
26. The component of velocity of image perpendicular to mirror is



$$\vec{V}_I = 2\vec{V}_m - \vec{V}_O$$
$$\Rightarrow (\vec{V}_1)_\perp = 2(-2) - (6) == 10 \text{ ms}^{-1}$$

For component of velocity of image parallel to the mirror
$(V_1)_| = 8 \text{ ms}^{-1}$
∴ velocity of image $(V_1) = \sqrt{(V_1)_\perp^2 + (V_1)^2}$
$\Rightarrow V_1 = \sqrt{100 + 64} = \sqrt{164} \text{ ms}^{-1}$ and $\theta = \tan^{-1}\left(\frac{4}{5}\right)$

27. $10\ ms^{-1}$



Velocity of object, $\vec{V}_O = (-5\sqrt{3}\hat{\imath} - 5\hat{\jmath})ms^{-1}$
Velocity of mirror, $\vec{V}_M = 3\hat{\imath}\ ms^{-1}$
For component of velocity perpendicular to mirror, we have
$(\vec{V}_{IM})_\perp = -(\vec{V}_{OM})_\perp = -(\vec{V}_O - \vec{V}_M)$
$\Rightarrow (\vec{V}_{IM})_\perp = -(-5\sqrt{3}\hat{\imath} - 3\hat{\imath}) = (5\sqrt{3}+3)\hat{\imath}ms^{-1}$
For component of velocity parallel to mirror, we have
$(\vec{V}_{IM})_\parallel = (\vec{V}_{OM})_\parallel = \vec{V}_O - \vec{V}_M = -5\hat{\jmath} - 0 = -5\hat{\jmath}$
Since, $(\vec{V}_{IM}) = (\vec{V}_{IM})_\perp + (\vec{V}_{IM})_\parallel$
$\Rightarrow (\vec{V}_{IM}) = (5\sqrt{3}+3)\hat{\imath} - 5\hat{\jmath}$

28. $\Rightarrow (\vec{V}_{IM}) = \vec{V}_1 - \vec{V}_M$
$\Rightarrow (5\sqrt{3}+3)\hat{\imath} - 5\hat{\jmath} = \vec{V}_1 - (3\hat{\imath})$
$\Rightarrow (5\sqrt{3}+3)\hat{\imath} + 3\hat{\imath} - 5\hat{\jmath} = \vec{v}_1$
$\vec{V}_I = (5\sqrt{3}+6)\hat{\imath} - 5\hat{\jmath}$
Further $\vec{v}_{1O} = \vec{V}_1 - \vec{v}_O$

29. $\vec{V}_{IO} = (5\sqrt{3}+6)\hat{\imath} - 5\hat{\jmath} - (-5\sqrt{3}\hat{\imath} - 5\hat{\jmath})$
$\vec{V}_{IO} = (5\sqrt{3}+6+5\sqrt{3})\hat{\imath} + (5-5)\hat{\jmath}$
$\vec{V}_{IO} = (10\sqrt{3}+6)\hat{\imath}ms^{-1}$

WORKSHEET – 4

Level – I

1) A	2) B	3) B	4) C	5) C	6) A	
7) D	8) A					

Level – II

1) D	2) D	3) D	4) B	5) C,D	6) D	7) D
8) D	9) A	10) D	11) A	12) A	13) A	14) C
15) A	16) B	17) A	18) a-4:b-3:c-1,2;d-4		19) A,B,C	20) A
21) D	22) A	23) C	24) A	25)A,D	26) A	27) B
28) A	29) B	30) A	31) A			

1. Inside the focus
2. i.e., object placed at centre of curvature.
   If object is at ' C ', image also forms at ' C '.
   Image distance = 2f.
3. X = Centre of curvature , Y = Focus , Z = pole
4. Object outside the focus
5. Virtual
6. Virtual, upright, and magnified

7. Convex mirrors have wide field of view.
8. The object is at the centre of curvature of the mirror.
9. concave mirrors concentrate sun rays at focus.
10. Half the radius of curvature 11) Anywhere in front of the mirror
11. A ray parallel to principal axis, appear to pass through focus after reflection from convex mirror.
12. Erect, magnified and virtual
13. Erect, diminished and virtual
14. Centre of curvature
15. Mirrors having curved reflecting surface are called as spherical mirrors.
16. spherical mirrors are of two types
17. Image position-Behind the mirror Image size-Diminished
    Image nature-Virtual
    Image forms - within focus
    a. Convex mirror acts as diverging mirror.
20. Image formed by convex mirror is virtual
21. For a concave mirror when object is placed at infinity the image is formed at principal focus.
22. Focal length(f)
23. Principal focus
24. Focal length = Radius of Curvature
    $f = \frac{F}{2} R = 2f = 2x$
23. $f = \frac{R}{2} R = 2x = 2(2x) = 4x$
24. $F = \frac{R}{2} R = 2f = 20$ cm
25. Convex mirrors have a wide field of view; Convex mirrors forms always erect image.
26. $R = yf = \frac{R}{2} = \frac{y}{2}$ cm
27. $f = \frac{x}{2} = \frac{40}{2} = 20$ cm
28. $f = \frac{R}{2} = \frac{2f}{2} = f$ cm
29. $f = \frac{R}{2} = \frac{12}{2} = 6$ cm
30. Spherical mirror (conceptual)
31. Focus, behind the mirror

WORKSHEET – 5

Level – I

1) A	2) B	3) B	4) B	5) A	6) B

Level – II

1) A	2) A	3) B	4) A	5) C	6) B	7) B
8) B	9) A	10) B	11) A	12) B	13) A	14) m=1
15) a-2;b-3;c-1;d-4		16) A	17) C	18) A	19) A	20) A
21) B	22) B	23) D	24) C	25) A	26) D	
27) B	28) A	29) D				

1. Convex, because convex mirror forms erect, virtual image and this mirror has large field of view.
2. R is negative
   $f = \frac{R}{2} = \frac{-15}{2} = -7.5$ cm
   Object distance = u = −10 cm

$\frac{1}{f}=\frac{1}{v}+\frac{1}{u};\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{-0.75}-\frac{1}{-10}=\frac{-1}{30};v=-30$ cm

3. $u=5$ cm
$\frac{1}{v}=\frac{1}{f}\frac{-1}{4}=\frac{1}{-7.5}-\frac{1}{-5}=\frac{1}{15};v=+15$ cm

4. $\mathrm{m}=\frac{-\mathrm{v}}{\mathrm{u}}=\frac{-(-30)}{-10}=-3$

5. $\mathrm{m}=\frac{-\mathrm{v}}{\mathrm{u}}=\frac{-15}{-5}=3$

6. $\frac{1}{v}+\frac{1}{u}=\frac{1}{f};\frac{1}{x_2}+\frac{1}{x_1}=\frac{1}{f};f=\frac{x_1x_2}{x_1+x_2}$

7. $\frac{1}{f}=\frac{2}{R}$

8. $f=\frac{R}{2}=\frac{-0.2}{2}=-0.1$ m

9. $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}\cdot m=\frac{-v}{u}\cdot -m=\frac{v}{u};\frac{v}{f}=\frac{v}{u}+\frac{v}{v};\frac{v}{f}=-m+1;m=1-\frac{v}{f}=\frac{f-v}{f}$

10. $f=-10$ cm$;u=8$ cm
$\frac{1}{f}=\frac{1}{4}+\frac{1}{v};-\frac{1}{10}=\frac{1}{8}+\frac{1}{v};\frac{1}{v}=\frac{-1}{10}+\frac{1}{8}=\frac{-4+5}{40};\frac{1}{v}=\frac{1}{40};v=40$ cm, virtual

11. Conceptual

12. Conceptual

13. $m=\frac{h_i}{h_0}$, conceptual

14. $m=\frac{h_i}{h_0};\ h_i=h_0;\ m=1$

15. Conceptual

16. Use $\frac{1}{\mathrm{f}}=\frac{1}{\mathrm{u}}+\frac{1}{\mathrm{v}}$
Where $f=-15$ cm and we get $u=-45$ cm
$-\frac{1}{15}=\frac{1}{45}+\frac{1}{v}$
$\frac{1}{V}=\frac{1}{45}-\frac{1}{15}$
$\frac{1}{\mathrm{V}}=\frac{1-3}{45}$
$v=22.5$ cm
$\mathrm{m}=\frac{\mathrm{v}}{\mathrm{u}}=-\frac{-22.5}{-45}=-\frac{1}{2}$
$\frac{\mathrm{h_i}}{\mathrm{h_0}}=\frac{1}{2}$
$\frac{\mathrm{h_i}}{5}=\frac{1}{2}$
$\mathrm{h_i}=2.5$ cm

17. Small and erect image is formed only by convex mirror. Plane mirror from images equal to object and concave mirror form images bigger than object.

18. Here, Object distance, $u=-15$ cm
focal length, $f=-5$ cm
Object height, $\mathrm{h_0}=0.2$ cm
We know, mirror formula,

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\Rightarrow V=\frac{uf}{u-f}$$

$$=\frac{(-15)(-5)}{-15+5}=-7.5 \text{ cm}$$

Again, magnification,

$$\mathrm{m}=-\frac{\mathrm{v}}{\mathrm{u}}=-\frac{-7.5}{-15}=-\frac{1}{2}$$

Now, $|\mathrm{m}|=\frac{\mathrm{h_i}}{\mathrm{h_o}}$

$\Rightarrow \mathrm{h_i}=|\mathrm{m}|\mathrm{h_0}=\frac{0.2}{2}$ cm = 0.1 cm

Thus, the image is formed at 7.5 cm from the pole of the mirror and its size is 0.1 cm. 19.

Here, Object distance, u = −30 cm

Focal length, $f = +20$ cm

We know, mirror formula, $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$\Rightarrow v=\frac{uf}{u-f}=\frac{-30\times 20}{-30-20}=+12$ cm

Again, magnification,

$\mathrm{m}=\frac{-\mathrm{v}}{\mathrm{u}}=\frac{-12 \text{ cm}}{30 \text{ cm}}=\frac{2}{5}$

Now, Image height = m × object height

$=\frac{2}{5}\times 0.5$ cm = 0.2 cm

Thus the image is formed behind the mirror at a distance of 12 cm from the pole. Image height is 0.2 cm.

20. $\mathrm{d_i}$ = 19.2 cm and Real

Use the equation $\frac{1}{f}=\frac{1}{d_o}+\frac{1}{d_i}$ where $f = 12$ cm and $d_o = 32$ cm.

21. $\mathrm{d_i}$ = 44 cm and f = 14.7 cm and Real

Use the equation $\frac{1}{f}=\frac{1}{d_o}+\frac{1}{d_i} w_{\text{h ere}}\ d_0 = 22$ cm.

If $M = 2$ and the image is inverted, then the d_i must be +44 cm. (It isn't-44 cm since the negative sign would only correspond to an upright and virtual image.) Solve for f one you find $\mathrm{d_i}$ = +44 cm.

22. $\frac{1}{\mathrm{f}}=\frac{-1}{\mathrm{u}}+\frac{1}{\mathrm{v}};\ \frac{1}{\mathrm{v}}=\frac{1}{\mathrm{f}}+\frac{1}{\mathrm{u}}=\frac{1}{\mathrm{f}}-\frac{1}{\mathrm{f}}=0.$

23. When object is at C. Image is also at C. minimum seperation between object \ and its real image is zero.

24. For both convex and concave mirrors.

25. A concave mirror can form a real as well as a virtual image for real object.

26. u = −50 cm, f = 25 cm

$$\frac{1}{25}=-\frac{1}{50}+\frac{1}{v};\frac{1}{v}=\frac{1}{25}+\frac{1}{50}=\frac{2+1}{50}=\frac{3}{50};v=\frac{50}{3}=16.6 \text{ cm}$$

27. Given u = −15 cm, f = −10 cm, O = 1 cm

$\frac{1}{v}+\frac{1}{u}=\frac{1}{f},\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{-10}-\frac{1}{-15}$

$\therefore v = -30$ cm

$$\frac{l}{0}=-\frac{v}{u}-\frac{-30}{-15}=-2; l=-2\times 1=-2 \text{ cm}$$

Image is inverted and on the same side (real) of size 2.

28. $m=-n; m=m=\frac{f}{f-u}; -n=\frac{-f}{-f-u}\Rightarrow nf+nu=-f$

$nu=-f-nf \Rightarrow u=\frac{-(n+1)}{n}f$

29. Here image can be real or virtual. If the image is real $f=-30, u=?, m=-3$

$m=\frac{f}{f-u}\Rightarrow -3=\frac{-30}{-30-u}; u=-40$ cm

If the image is virtual

$m=\frac{f}{f-u}\Rightarrow 3=\frac{-30}{-30-u}\Rightarrow u=-20$ cm

WORKSHEET – 6

Level – I

1) 1	2) 1	3) 3	4) 4	5) 3	6) 2	7) 1
8) 2	9) 3	10) n=4	11) 1	12) 3	13) 3	14) 1
15) 1						

1. consider reflection at concave mirror.

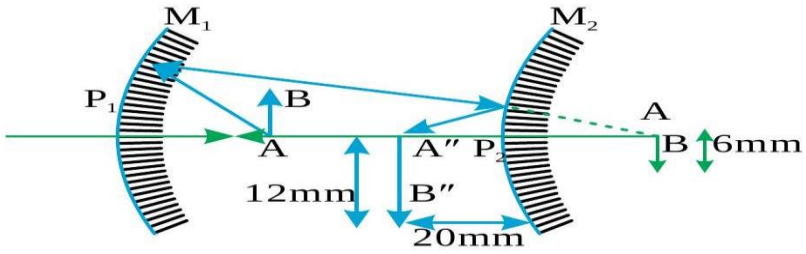
$(M_1) u=-20$ cm, $f=-15$ cm using mirror equation we get,

$v=\frac{uf}{(u-f)}=[(-20)(-15)]/[(-20)-(-15)]=-60$ cm

$m_1=\frac{-v}{u}=-\frac{(-60)}{-20}=-3$(inverted)

$A'B'=m_1(AB)=6$ mm

The image ($A'B'$) formed by concave mirror acts as object for convex mirror. Now, consider reflection at convex mirror (M_2).



u = +10 cm, f = +20 cm

$$v=\frac{vf}{(u-f)}=[(10)(20)]/[10-20]=-20 \text{ cm}$$

2. Image is real.

3. $m_2=v=\frac{-v}{u}=\frac{-20}{10}=2$(Erect)

$A''B''=m_2(AB)=12$ mm

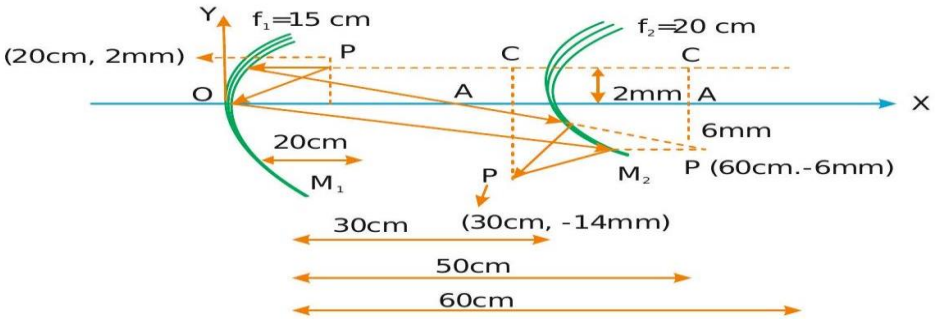
Hence, for final image A "B"

Position : 20 cm in front of convex mirror (M_2)

Nature : Real and inverted

Size : 12 mm

4.



For reflection at concave mirror M_1

$u = -20$ cm, $f_1 = -15$ cm

$$v_1 = u\frac{f_1}{(u - f_1)} = [(-20)(-15)]/[-20 + 15] = -60 \text{ cm}$$

5. Magnification $(m_1) = -v_1/u = -\left[\frac{(-60)}{(-20)}\right] = -3$ (Inverted)
6. Image is real and inverted
7. $A'P' = m_1(AP) = 3 * 2 = 6$ mm

For reflection at convex mirror M_2

$u = +10$ cm, $f_1 = +20$ cm

$$v_2 = \frac{uf_2}{(u - f_2)} = [(10)(20)]/[10 - 20] = -20 \text{ cm}$$

8. Magnification $(m_2) = -\frac{v_2}{u} = -\left[\frac{(-20)}{(10)}\right] = 2$

When S is placed at the centre of curvature, the image will form at the ssame point, so in this case the value of d will be $2f + 2f = 4f$

$nf = 4f$

$n = 4$.

11. For mirror M_1: $u = 30$ cm, $f = -20$ cm

By mirror formula, $\left(\frac{1}{u}\right) + \left(\frac{1}{v}\right) = \left(\frac{1}{f}\right)$,

we have $\left(\frac{1}{-30}\right) + \left(\frac{1}{v}\right) = \left(\frac{1}{-30}\right)$

which on solving gives, $v = 60$ cm

For mirror M_2 The image formed by mirror M_1 is a virtual object for mirror M_2. Let it is a distance x from the pole of mirror M_2.

Thus $u_2 = +x, v_2 = -(30 - x)$

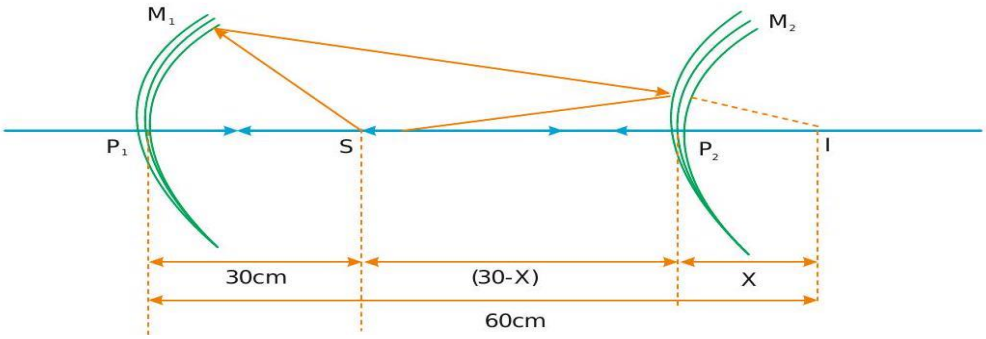
Again by mirror formula. We have

$$\left(\frac{1}{x}\right) + \left(\frac{1}{-(30 - x)}\right) = \left(\frac{1}{20}\right)$$

which on solving gives $x = 10$ cm or 60 cm, $x = 60$ cm is not possible, thus $x = 10$ cm.

Thus the separation between the mirrors $= 60 - 10 = 50$ m

12. The image formed by mirror M_2 is at a distance 10 cm



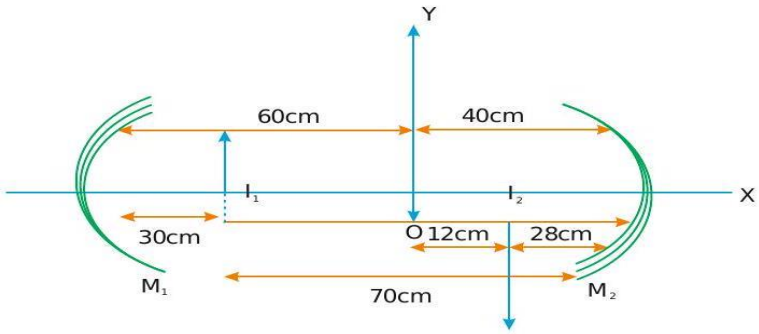
13. The distance of the object from the plane mirror is 30 cm and so the distance of its image is also 30 cm from the mirror. As image formed by both the mirrors coincide, so distance of image for convex mirror is = 10 cm.

By mirror formula $\left(\frac{1}{u}\right)+\left(\frac{1}{v}\right)=\left(\frac{1}{f}\right);\left(\frac{1}{f}\right)=\left(\frac{1}{10}\right)+\left(\frac{1}{-50}\right)$

Which on solving give $f = 12.5$ cm

$\therefore$ Radius of curvature R = 2f = 25 cm

14.



Using mirror formula for first reflection :

$\left(\frac{1}{f}\right)=\left(\frac{1}{v}\right)+\left(\frac{1}{u}\right)\Rightarrow\left(\frac{1}{-20}\right)=\left(\frac{1}{v}\right)+\left(\frac{1}{-70}\right)\Rightarrow\left(\frac{1}{v}\right)=\left(\frac{1}{60}\right)-\left(\frac{1}{20}\right)\Rightarrow v = 30$ cm

Using mirror formula for first reflection :

$\left(\frac{1}{f}\right)=\left(\frac{1}{70}\right)-\left(\frac{1}{20}\right)=\left[\frac{(2-7)}{140}\right]$

$\Rightarrow \mathrm{v}=-\left(\frac{140}{5}\right)=-28$ cm

Height of $\mathrm{I}_2 \Rightarrow \mathrm{m}=\left(\frac{30}{-60}\right)=1=\mathrm{I}_1=\left(\frac{1}{2}\right)$ cm

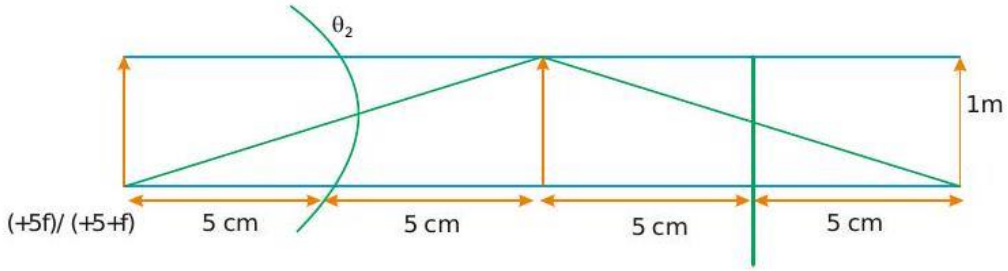
Height of first image from x-axis $= 1+\left(\frac{1}{2}\right)=\frac{3}{2}$ cm

Height of $\mathrm{I}_2 \Rightarrow \mathrm{m}=\left(\frac{-28}{-70}\right)=\frac{21_2}{3}$

$\Rightarrow I_2=\frac{[3\times 28]}{2\times 70}$

$\mathrm{I}_2 = -0.6$ cm co-ordinate of $\mathrm{I}_2 = (12-0.6)$

15. $\frac{1}{10}=\theta_1 = 0.1$rad.



$\left(\frac{h_1}{1}\right)=\frac{5f}{(5+f)}\Rightarrow h_1=\frac{f}{(5+f)}$

$\theta_2 = [f/(5+f)]/5 + [5f/(5+f)] = f/[25+10f] = 0.1/2$

$2f = 2.5 + f$

f = 2.5 m